Math 210: Real Analysis
Spring 2025: Lectures given by A. Sabra
American University of Beirut

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Chapter 1

Set Theory, The set of Integers and Cardinality

In this chapter, we will introduce set theoretic notation, contruct the set of integers, and discuss the concept of cardinality i.e the size of sets.

Motivation 1.1. The Hilbert Paradox: Consider a hotel with infinitely many rooms $\{1, 2, 3, \dots\}$, all of which are occupied by guests $\{a_1, a_2, a_3, \dots\}$. We assign guest a_i to room i so the hotel is full and all guests are served. Now suppose a new guest x^* arrives, can we serve this guest? Yes, we can! We can move guest a_1 to room 2, guest a_2 to room 3, and in general guest a_i to room $i+1$. This way, we free up room 1 for the new guest x^*. So this is as if I am saying that $\{a_1, a_2, \dots\}$ and $\{x^*, a_1, a_2\}$ have the same size. So we have to define what "size" means for finite and infinite sets and we need to be consistent with this definition.

1.1 Operations on Sets

Definition 1.1. In naive set theory, a set is a collection of elements. So given an element x and a set A, we either have :

- $x \in A$ (i.e. x is an element of A) or
- $x \notin A$ (i.e. x is not an element of A).

Definition 1.2. Given 2 sets A and B, we say that A is a subset of B if and only if every element $x \in A$ we have $x \in B$. In this case $A \subseteq B$

$$A \subseteq B \iff x \in A \implies x \in B \iff x \notin B \implies x \notin A$$

Moreover if $A \subseteq B$ and $B \subseteq A$ then the sets A and B are said to be equal, and we write $A = B$

Definition 1.3. $\emptyset$ denotes the empty set, that is the set that contains no element. Notice that for any set A we have $\emptyset \subseteq A$.

Definition 1.4. Give a set A, the power set of A denoted by $\mathcal{P}(A)$ is the set of all subsets of A that is $B \in \mathcal{P}(A) \iff B \subseteq A$

Definition 1.5. Given 2 sets A and B we define:

(1) The union $A \cup B = \{x : x \in A \text{ or } x \in B\}$

(2) The intersection $A \cap B = \{x : x \in A \text{ and } x \in B\}$

(3) The complement of B relative to A is $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$

(4) Let $A \subseteq X$, the complement of A relative to X is $X \setminus A = A^c = \{x : x \in X \text{ and } x \notin A\}$

Example 1.1. Let $A = \{a, b\}$ and $B = \{b, c\}$ then we have:

- $A \cup B = \{a, b, c\}$
- $A \cap B = \{b\}$
- $A \setminus B = \{a\}$
- $B \setminus A = \{c\}$

Definition 1.6. We say that A and B are disjoint if and only if $A \cap B = \emptyset$.

Lemma 1.1. *Given A and B sets we have:*

(1) $A \subseteq A \cup B$

(2) $A \cap B \subseteq A$

Proof. (1) $x \in A \implies x \in A \cup B$ since $x \in A$ or $x \in B$.

(2) $x \in A \cap B \implies x \in A$ since $x \in A$ and $x \in B$.

□

Theorem 1.1 (De Morgan's Laws)**.** *Given A and B sets we have:*

(1) $(A \cup B)^c = A^c \cap B^c$

(2) $(A \cap B)^c = A^c \cup B^c$

Proof of 1. To prove set equality, we show both inclusions:

$$(A \cup B)^c \subseteq A^c \cap B^c \quad \text{and} \quad A^c \cap B^c \subseteq (A \cup B)^c.$$

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \cup B \\ &\iff (x \notin A) \text{ and } (x \notin B) \\ &\iff (x \in A^c) \text{ and } (x \in B^c) \\ &\iff x \in A^c \cap B^c, \end{aligned}$$

so $(A \cup B)^c \subseteq A^c \cap B^c$.

$$\begin{aligned} y \in A^c \cap B^c &\iff (y \in A^c) \text{ and } (y \in B^c) \\ &\iff (y \notin A) \text{ and } (y \notin B) \\ &\iff y \notin A \cup B \\ &\iff y \in (A \cup B)^c, \end{aligned}$$

so $A^c \cap B^c \subseteq (A \cup B)^c$.

Hence, $(A \cup B)^c = A^c \cap B^c$. The proof of the second part is similar and left as an exercise. □

Theorem 1.2 (Distributive Laws)**.** *Given A, B and C sets we have:*

(1) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(2) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Definition 1.7. Given 2 sets A and B, we define the *Cartesian Product* to be

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

Example 1.2. Let $A = \{1, 2\}$ and $B = \{3, 4\}$ then we have:

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

1.1.1 Functions on sets

Definition 1.8. A relation f between X and Y is a subset of $X \times Y$ ($f \subseteq X \times Y$). That is every element of the power set $\mathcal{P}(X \times Y)$ is a relation from X to Y.

Definition 1.9. A function f between X and Y is a relation from X to Y such that for every $x \in X$ there exists exactly one element $y \in Y$ such that $(x, y) \in f$. We write:

$$f : X \to Y \text{ and } f(x) = y \iff (x, y) \in f$$

We call X the domain of f and $f(X)$ the range of f

Example 1.3. Give $A = \{\alpha, \beta, \gamma\}$ and $B = \{a, b\}$:

- $\{(\alpha, a), (\beta, b), (\gamma, a), (\alpha, b)\}$ is a relation but not a function since α is related to both a and b.
- $\{(\alpha, a), (\beta, b), (\gamma, a)\}$ is a function since every element of A is related to exactly one element of B
- $\{(\alpha, a), (\beta, b)\}$ is not a function since γ is not related to any element of B.

Definition 1.10. Given 2 sets X and Y and a function $f : X \to Y$

- For $E \subseteq X$, the image of E under f is $f(E) = \{y \in Y : y = f(x) \text{ for some } x \in E\}$
- For $H \subseteq Y$, the preimage of H under f is $f^{-1}(H) = \{x \in X : f(x) \in H\}$.

Remark 1.1. Let $f : X \to Y$ be a function:

(1) If $A, B \subseteq X$ such that $A \subseteq B$ then $f(A) \subseteq f(B)$

(2) If $H, K \subseteq Y$ such that $H \subseteq K$ then $f^{-1}(H) \subseteq f^{-1}(K)$

Proof.

- Let $y \in f(A)$ then there exists $x \in A$ such that $f(x) = y$. Since $A \subseteq B$ we have $x \in B$ and thus $y = f(x) \in f(B)$ so $f(A) \subseteq f(B)$
- Let $x \in f^{-1}(H)$ then $f(x) \in H$. Since $H \subseteq K$ we have $f(x) \in K$ and thus $x \in f^{-1}(K)$ so $f^{-1}(H) \subseteq f^{-1}(K)$

□

Example 1.4. Let $X = \{\alpha, \beta, \gamma\}$ and $Y = \{a, b\}$ and we define $f : X \to Y$ such that:

$$f(\alpha) = a, f(\beta) = b, f(\gamma) = a$$

Then we have:

- $f^{-1}(\{a\}) = \{\alpha, \gamma\}$
- $f^{-1}(\{b\}) = \{\beta\}$

In general: if $f : X \to Y$ and $E \subseteq X$, then $f^{-1}(f(E))$ is not necessarily equal to E.

Claim. Let $f : X \to Y$ and $E \subseteq X$ then $f^{-1}(f(E)) \supseteq E$.

Proof. Let $x \in E \implies f(x) \in f(E)$ so $x \in f^{-1}(f(E))$ and thus $E \subseteq f^{-1}(f(E))$. □

Definition 1.11. Let $f : X \to Y$ be a function, we say that f is injective or one-to-one if and only if whenever $x_1 \neq x_2$ we have $f(x_1) \neq f(x_2)$ or equivalently if $f(x_1) = f(x_2)$ then $x_1 = x_2$.
Let $H = f(X)$ be the range of f. In this case, every element of H has only one preimage. We identify the map f^{-1} by the function $f^{-1} : H \to X$ such that $f^{-1}(y) = x \iff f(x) = y$

Definition 1.12. Let $f : X \to Y$ be a function, we say that f is surjective or onto if and only if $f(X) = y$, that is every image in Y has at least one preimage in X.

Definition 1.13. Let $f : X \to Y$ be a function, we say that f is bijective if and only if f is both injective and surjective. In this case, every element of Y has exactly one preimage in X and we can identify the map f^{-1} by the function $f^{-1} : Y \to X$ such that $f^{-1}(y) = x \iff f(x) = y$

Exercise 1.1. $f : X \to Y$ function suc that $f(f^{-1}(H)) \subseteq H \quad \forall H \subseteq Y$ adn show that $f(f^{-1}(H)) = H$ if and only if f is surjective.

Exercise 1.2. Let $f : X \to Y$ be a function. Show that $f(A \cup B) = f(A) \cup f(B)$ and $f(A \cap B) \subseteq f(A) \cap f(B)$ for all $A, B \subseteq X$. Show that $f(A \cap B) = f(A) \cap f(B)$ for all $A, B \subseteq X$ if and only if f is injective.

Exercise 1.3. Let $f : X \to Y$ be a function. Show that $f^{-1}(H \cap K) = f^{-1}(H) \cap f^{-1}(K)$ for all $H, K \subseteq Y$, and give an example in which the inclusion $f(A \cap B) \subseteq f(A) \cap f(B)$ is proper and one in which it isn't.

1.2 The set of Natural Numbers

Definition 1.14 (Peano Axioms)**.** The set of natural numbers $\mathbb{N}$ is the smallest set such that:

- $1 \in \mathbb{N}$
- If $n \in \mathbb{N}$ then $n++ \in N$
- 1 doesn't have a predecessor
- If $n++ = m++$ then $n = m$
- Principle of Mathematical Induction on $S \subseteq \mathbb{N}$: If $1 \in S$ and if $n \in S$ implies $n++ \in S$ then $S = \mathbb{N}$

Definition 1.15. We define addition recursively on $\mathbb{N}$ as follows:

- $n + 1 = n++$
- $n + (m++) = (n + m)++$

Proposition 1.1. *If $n \in \mathbb{N}$ then $n = 1$ or $n = m++$ for some $m \in \mathbb{N}$*

Proof. We use induction, $P(n) = \{n = 1 \text{ or } n = m++ \text{ for some } m \in \mathbb{N}\}$.

- **Base Case:** P(1) is true since $1 = 1$.
- **Induction Step:** Assume $P(n)$ is true for some $n \in \mathbb{N}$ then $n = m++$ for $m \in \mathbb{N}$ then $n++ = (m++)++$ and $(m++)++ \in \mathbb{N}$ so $P(n)$ is true.

□

Proposition 1.2. $b + a = c + a \implies b = c$ *for all* $a, b, c \in \mathbb{N}$.

Proof. We use induction on a.

- **Base Case:** Let $a = 1$ then $b + 1 = c + 1$ implies $b++ = c++$ and thus $b = c$.
- **Induction Step:** Assume the statement is true for some $a \in \mathbb{N}$ then let $a++$ be the successor of a. We have $b + (a++) = c + (a++)$ implies $(b + a)++ = (c + a)++$ and thus $b + a = c + a$ and by the induction hypothesis we have $b = c$.

□

Lemma 1.2. $1 + n = n + 1 = n++$ *for all* $n \in \mathbb{N}$.

Proof. We use induction on n.

- **Base Case:** Let $n = 1$; then $1 + 1 = 1++$.
- **Induction Step:** Assume $1 + n = n++$ for some $n \in \mathbb{N}$. Then $1 + (n++) = (1 + n)++ = (n++)++$.

□

Lemma 1.3. $(n + 1) + m = n + (m + 1) = (n + m) + 1$ *for all* $n, m \in \mathbb{N}$.

Proof. We use induction on m.

- **Base Case:** Let $m = 1$ then $(n + 1) + 1 = n + (1 + 1) = (n + 1) + 1$ and thus the statement is true.
- **Induction Step:** Assume the statement is true for some $m \in \mathbb{N}$ then let $m++$ be the successor of m. We have $(n + 1) + (m++) = ((n + 1) + m)++ = ((n + m) + 1)++ = (n + (m++)) + 1$ and thus $(n + 1) + (m++) = n + ((m++) + 1) = (n + (m++)) + 1$.

□

Proposition 1.3. $n + m = m + n$ *for all* $n, m \in \mathbb{N}$.

Proof. We use induction on m.

- **Base Case:** Let $m = 1$ then $n + 1 = 1 + n$ and thus the statement is true.
- **Induction Step:** Assume the statement is true for some $m \in \mathbb{N}$ then let $m++$ be the successor of m. We have $n + (m++) = (n + m)++ = (m + n)++ = (m++) + n$ and thus $n + (m++) = (m++) + n$.

□

Proposition 1.4. *$(a+b)+c = a+(b+c)$ for all $a, b, c \in \mathbb{N}$.*

Proof. We use induction on c.

- **Base Case:** Let $c = 1$ then $(a+b)+1 = a+(b+1)$ and thus the statement is true.
- **Induction Step:** Assume the statement is true for some $c \in \mathbb{N}$ then let $c++$ be the successor of c. We have $(a+b)+(c++) = ((a+b)+c)++ = (a+(b+c))++ = a+((b+c)++)$ and thus $(a+b)+(c++) = a+(b+(c++))$.

□

Definition 1.16. We define multiplication recursively on $\mathbb{N}$ as follows:

- $n \cdot 1 = n$
- $n \cdot (m++) = n \cdot m + n$

Lemma 1.4. *$(a+b)c = ac+bc$ for all $a, b, c \in \mathbb{N}$.*

Proof. We use induction on c.

- **Base Case:** Let $c = 1$ then $(a+b)1 = a1+b1$ and thus the statement is true.
- **Induction Step:** Assume the statement is true for some $c \in \mathbb{N}$ then let $c++$ be the successor of c. We have $(a+b)(c++) = (a+b)c+(a+b) = ac+bc+a+b = ac+a+bc+b = a(c++)+b(c++)$ and thus $(a+b)(c++) = ac+a+bc+b = a(c++)+b(c++)$.

□

Lemma 1.5. *$1 \cdot n = n \cdot 1 - n$ for all $n \in \mathbb{N}$.*

Proof. Left as an exercise. □

Consequence 1.1. *$2 = 1++$ and $2c = (1+1)c = 1c+1c = c+c$ for all $c \in \mathbb{N}$. Recursively we get $mn = n+n+\cdots+n$ (m times) for all $m, n \in \mathbb{N}$.*

Definition 1.17. We define exponentiation recursively on $\mathbb{N}$ as follows:

- $n^1 = n$
- $n^{m++} = n^m \cdot n$

Lemma 1.6. *We get the following properties of exponentiation:*

- *$n^{m+k} = n^m \cdot n^k$ for all $n, m, k \in \mathbb{N}$.*
- *$(n^m)^k = n^{mk}$ for all $n, m, k \in \mathbb{N}$.*
- *$(nm)^k = n^k m^k$ for all $n, m, k \in \mathbb{N}$.*

1.3 Orders:

Definition 1.18. Given a set S, a (strict) order on S is a relation from S to S such that:

(1) Trichotomy: $\forall a, b \in S$ exactly one of the following holds: $a < b$, $a = b$, or $a > b$.

(2) Transitivity: $\forall a, b, c \in S$ if $a < b$ and $b < c$ then $a < c$.

We say that S is an ordered set if there exists an order on S.

Example 1.5. $S = \{a, b, c\}$ and we define the order $<$ on S such that $a < b < c$ then S is an ordered set.

Definition 1.19. Given $a, b \in S$

- $a < b \iff b > a$
- $a \leq b \iff a < b$ or $a = b$
- $a \geq b \iff a > b$ or $a = b$

Theorem 1.3. *"Succeeds" is an order relation on $\mathbb{N}$ we say $a < b$ if and only if b succeeds a that is $b = a++$ or $b = (a++)++$ or ....*

Proof. We prove the two defining properties of a strict order.

(1) **Trichotomy.** We argue by induction on b.

Let $P(b)$ be the statement: for every $a \in \mathbb{N}$, exactly one of

$$a = b, \qquad a < b, \qquad b < a$$

holds.

Base case: $b = 1$. By the proposition proved earlier, every $a \in \mathbb{N}$ is either 1 or a successor of some natural number. Hence either $a = 1$, or a is obtained from 1 by finitely many applications of the successor map, in which case $1 < a$.

Induction step: Assume $P(b)$ holds. We prove $P(b++)$. Let $a \in \mathbb{N}$.

- If $a = b++$, we are done.
- If $a < b$, then by definition of the order, a succeeds to b in finitely many steps, and since $b < b++$, we get $a < b++$.
- If $b < a$, then either $a = b++$ or $b++ < a$; in either case, the trichotomy for a and $b++$ holds.

Thus exactly one of $a = b++$, $a < b++$, or $b++ < a$ holds.

Therefore $P(b)$ is true for all $b \in \mathbb{N}$.

(2) **Transitivity.** Suppose $a < b$ and $b < c$. By the definition of the order, there exist $m, n \in \mathbb{N}$ such that

$$b = a + m \qquad \text{and} \qquad c = b + n.$$

Using associativity of addition proved earlier,

$$c = (a + m) + n = a + (m + n).$$

Since $m + n \in \mathbb{N}$, it follows that $a < c$.

Hence the relation "Succeeds" is a strict order on $\mathbb{N}$. □

Exercise 1.4. Show that if $a + c < b + c$ then $a < b$ for all $a, b, c \in \mathbb{N}$. and if $ac < bc$ then $a < b$ for all $a, b, c \in \mathbb{N}$ such that $c \neq 0$, and $ac = bc$ implies $a = b$ for all $a, b, c \in \mathbb{N}$ such that $c \neq 0$.

Definition 1.20 (Strong Induction)**.** Let $S \subseteq \mathbb{N}$ such that:

- $1 \in S$
- If $1, 2, \ldots, n \in S$ then $n++ \in S$

then $S = \mathbb{N}$.

Proof. Let $T = \{n \in S : \{1, \ldots n\} \subseteq S\}$, $T \subseteq S$

- **Base Case:** $1 \in T$ since $\{1\} \subseteq S$.
- **Induction Step:** Assume $n \in T$ then $\{1, \ldots n\} \subseteq S$ and thus $n++ \in S$ and thus $\{1, \ldots n, n++\} \subseteq S$ and thus $n++ \in T$.

By induction, $T = \mathbb{N}$ and $T \subseteq S \subseteq \mathbb{N}$ so $S = \mathbb{N}$ and thus the principle of strong induction is true. □

Corollary 1.1. *Given a sequence of statements $P(n)$ indexed by $\mathbb{N}$ such that $P(1)$ is true and if $P(1), P(2), \ldots, P(n)$ are true then $P(n++)$ is true then $P(n)$ is true for all $n \in \mathbb{N}$.*

Proof. $S = \{n \in \mathbb{N} : P(n) \text{ is true } \}$. □

Corollary 1.2. *Given $n, m \in \mathbb{N}$ we say that m divides n if and only if there exists $k \in \mathbb{N}$ such that $n = mk$. We say p is prime if and only if $p \geq 2$ and the only divisors of p are 1 and p. Then every natural number $n \geq 2$ is either prime or can be written as a product of primes.*

Proof. We use induction on n.

- **Base Case:** Let $n = 2$ then 2 is prime and thus the statement is true.
- **Induction Step:** Assume the statement is true for all $2 \leq k < n$ then if n is prime we are done. If n is not prime then there exists $a, b \in \mathbb{N}$ such that $n = ab$ and $1 < a, b < n$. By the induction hypothesis, a and b are either prime or can be written as a product of primes. Thus, $n = ab$ can be written as a product of primes.

□

Theorem 1.4 (Well Ordering Principle)**.** *Every non-empty subset of $\mathbb{N}$ has a smallest element. That is if $S \subseteq \mathbb{N}$ and $S \neq \emptyset$ then there exists $p \in S$ such that $p \leq n$ for all $n \in S$. We denote this by $p = \min(S)$.*

Proof. Let $P(n)$ be the statement that if $S \cap \{1, 2, \ldots, n\} \neq \emptyset$, then this set has a smallest element.

- **Base Case:** For $n = 1$, if $1 \in S$ then $S \cap \{1\}$ has smallest element 1.
- **Induction Step:** Assume $P(n)$ is true for some $n \in \mathbb{N}$. If $S \cap \{1, 2, \ldots, n++\} \neq \emptyset$, then either $S \cap \{1, 2, \ldots, n\} \neq \emptyset$, in which case it has a smallest element by the induction hypothesis, or else $S \cap \{n++\} \neq \emptyset$, in which case $n++$ is the smallest element.

Therefore $P(n)$ is true for all $n \in \mathbb{N}$, and every non-empty subset of $\mathbb{N}$ has a smallest element. □

Remark 1.2. Induction $\implies$ Strong Induction $\implies$ Well Ordering Principle. Remains to show that Well Ordering Principle $\implies$ Induction.

Proof. Let S be a set such that:

(1) $1 \in S$

(2) If $n \in S$ then $n++ \in S$

We want to show that $S = \mathbb{N}$. Assume by contradiction that $S \neq \mathbb{N}$ then $\mathbb{N} \setminus S \neq \emptyset$ and thus by the well ordering principle there exists $p = \min(\mathbb{N} \setminus S)$. Since $1 \in S$ we have $p \neq 1$ and thus $p - 1$ is a natural number. Since p is the smallest element of $\mathbb{N} \setminus S$ we have $p - 1 \in S$ and thus $p = (p-1)++ \in S$ which is a contradiction since $p \in \mathbb{N} \setminus S$. Thus, $S = \mathbb{N}$ and the principle of induction is true. □

1.4 Equivalence Relations: Construction of the integers

Definition 1.21. Given a set A an equivalence relation on A is a relation R from A to A such that:

(1) Reflexivity: $\forall a \in A$ we have aRa

(2) Symmetry: $\forall a, b \in A$ if aRb then bRa

(3) Transitivity: $\forall a, b, c \in A$ if aRb and bRc then aRc

Example 1.6. $\mathbb{N} \times \mathbb{N} = \{(m, n) : m, n \in \mathbb{N}\}$ then $(a, b) \sim (c, d)$ if and only if $a = c$ and $b = d$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Definition 1.22.

Definition 1.23. Given an equivalence relation R on a set A and $a \in A$ we define the equivalence class of a to be $[a] = \{x \in A : xRa\}$.

Definition 1.24. Given an equivalence relation R on a set A and $a \in A$ we define the equivalence class of a to be $[a] = \{x \in A : xRa\}$.

We define an equivalence relation on $\mathbb{N} \times \mathbb{N}$ by $(a, b) \sim (c, d)$ if and only if $a + d = b + c$. We denote the equivalence class of (a, b) by $[(a, b)]$ or $\overline{(a, b)}$.

Proposition 1.5. *The relation $\sim$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.*

Proof. We verify the three properties:

(1) **Reflexivity:** $(a, b) \sim (a, b)$ since $a + b = b + a$ by commutativity of addition.

(2) **Symmetry:** If $(a, b) \sim (c, d)$ then $a + d = b + c$, so $c + b = d + a$ and thus $(c, d) \sim (a, b)$.

(3) **Transitivity:** If $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$ then $a + d = b + c$ and $c + f = d + e$. Adding these equations, $(a + d) + (c + f) = (b + c) + (d + e)$, which gives $a + c + d + f = b + d + c + e$ and thus $a + f = b + e$, so $(a, b) \sim (e, f)$.

□

Definition 1.25. The set of integers is defined as

$$\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\sim = \{[(a, b)] : a, b \in \mathbb{N}\}$$

where $[(a, b)]$ denotes the equivalence class of (a, b) under $\sim$.

Example 1.7. $[(2,1)] = [(3,2)] = [(4,3)] = \dots$ represents the integer 1.
$[(1,2)] = [(2,3)] = [(3,4)] = \dots$ represents the integer -1.
$[(n,n)]$ for all $n \in \mathbb{N}$ represents the integer 0.

Definition 1.26. We define addition as follows:

$$[(a,b)] + [(c,d)] = [(a+c, b+d)]$$

Is this well defined? That is if $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$ then do we have $(a+c, b+d) \sim (a'+c', b'+d')$? Yes, since $a+d = b+c$ and $a'+d' = b'+c'$ then $(a+c)+(b'+d') = (a+b')+(c+d') = (b+a')+(d+c') = (b+d)+(a'+c')$ and thus $(a+c, b+d) \sim (a'+c', b'+d')$.

Claim. We claim the following:

(1) $[n] + [m] = [n+m]$ for all $n, m \in \mathbb{N}$

(2) $[0] + [n] = [n]$ for all $n \in \mathbb{N}$

(3) $[n] + [-n] = [0]$ for all $n \in \mathbb{N}$

Proof.

- $[n] + [m] = [(n,0)] + [(m,0)] = [(n+m, 0+0)] = [(n+m, 0)] = [n+m]$.
- $[0] + [n] = [(0,0)] + [(n,0)] = [(0+n, 0+0)] = [(n,0)] = [n]$.
- $[n] + [-n] = [(n,0)] + [(0,n)] = [(n+0, 0+n)] = [(n,n)] = [0]$.

□

Definition 1.27. Define multiplication as follows:

$$[(a,b)] \cdot [(c,d)] = [(ac+bd, ad+bc)]$$

As an exercise check if this is well defined.

Claim. We claim the following:

(1) $[n] \cdot [m] = [nm]$ for all $n, m \in \mathbb{N}$

(2) $[0] \cdot [n] = [0]$ for all $n \in \mathbb{N}$

(3) $[1] \cdot [n] = [n]$ for all $n \in \mathbb{N}$

(4) $[-1] \cdot [n] = [-n]$ for all $n \in \mathbb{N}$

(5) $[-1] \cdot [-n] = [n]$ for all $n \in \mathbb{N}$

Proof. — Left as an exercise. □

Definition 1.28. Define $[a] - [b] = [a] + [-1][b]$

Definition 1.29. We define the order on $\mathbb{Z}$ as follows:

$$[(a,b)] < [(c,d)] \iff a+d < b+c$$

We need to check if this is well defined. That is if $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$ then do we have $a+d < b+c$ if and only if $a'+d' < b'+c'$? Yes, since $a+d = b+c$ and $a'+d' = b'+c'$ then $a+d < b+c$ if and only if $b+c < a+d$ if and only if $b'+c' < a'+d'$ if and only if $a'+d' < b'+c'$.

Definition 1.30. We define $\mathbb{Q} = \mathbb{Z} \times \mathbb{N}$ and we define an equivalence relation $\sim$ on $\mathbb{Q}$ by $(a,b) \sim (c,d)$ if and only if $ad = bc$. We denote the equivalence class of (a,b) by $[(a,b)]$ or $\frac{a}{b}$, and the construction of $\mathbb{Q}$ is left as the group work.

1.5 Countability

Definition 1.31. $\mathbb{N}_n = \{1, 2, \ldots, n\}$ for all $n \in \mathbb{N}$.

Theorem 1.5 (Pigeonhole Principle)**.** *If $m > n$ there does not exist an injection from $\mathbb{N}_m$ to $\mathbb{N}_n$.*

Proof. It is enough to prove that for every $n \in \mathbb{N}$ there is no injection from N_{n+1} to N_n.

We argue by induction on n.

Base case: For $n = 1$, there is no injection from N_2 to N_1, since two distinct elements would have to have distinct images in a one-element set.

Induction step: Assume there is no injection from N_{n+1} to N_n. Suppose, for contradiction, that

$$f : N_{n+2} \to N_{n+1}$$

is injective.

If $n+1$ is not in the range of f, then in fact $f(N_{n+2}) \subseteq N_n$, and restricting f to $N_{n+1} \subset N_{n+2}$ gives an injection from N_{n+1} to N_n, contradicting the induction hypothesis.

If $n+1$ is in the range of f, let j be the unique element of N_{n+2} such that $f(j) = n+1$. Define $g : N_{n+1} \to N_n$ by

$$g(i) = \begin{cases} f(i), & i < j, \\ f(i+1), & i \geq j. \end{cases}$$

Then g is injective, again contradicting the induction hypothesis.

So no injection from N_{n+2} to N_{n+1} exists.

Therefore, for any $m > n$, an injection $N_m \to N_n$ would restrict to an injection $N_{n+1} \to N_n$, which is impossible. Hence no such injection exists.

□

Definition 1.32. We say that a set A is finite if and only if $A = \emptyset$ or there exists a bijection from A to $\mathbb{N}_n$ for some $n \in A$ otherwise we say that A is infinite. For a finite set A we have

$$|A| = \begin{cases} 0 & \text{if } A = \emptyset \\ n & \text{if there exists a bijection from } A \text{ to } \mathbb{N}_n \end{cases}$$

The cardinality is well defined. Suppose there exist two bijections $f : A \to \mathbb{N}_n$ and $g : A \to \mathbb{N}_m$ then $g \circ f^{-1} : \mathbb{N}_n \to \mathbb{N}_m$ is a bijection and thus by the pigeonhole principle we have $n = m$ and thus the cardinality of A is well defined.

Theorem 1.6 (Pigeonhole infinite edition)**.** $\mathbb{N}$ *is an infinite set.*

Proof. $N \neq \emptyset$. Suppose that N is finite then there exists a bijection from $\mathbb{N}$ to $\mathbb{N}_n$ for some $n \in \mathbb{N}$. This is a contradiction since there is no injection from $\mathbb{N}$ to $\mathbb{N}_n$ by the pigeonhole principle. Thus, $\mathbb{N}$ is an infinite set. □

Definition 1.33. We say that A is countably infinite if and only if there exists a bijection from A to $\mathbb{N}$, we denote $|A| = \aleph_0$. We say A is countable if and only if A is finite or countably infinite, otherwise we say that A is uncountable.

Proposition 1.6. *$A \subseteq \mathbb{N}$ then A is countable. That is $\aleph_0$ is the smallest infinity.*

Proof. If A is finite then we are done. If A is infinite then let n_1 be the minimum of A by well ordering principle and let n_2 be the minimum of $A \setminus \{n_1\}$ and so on we get a sequence of natural numbers $n_1 < n_2 < \dots$ and thus we can define a bijection from $\mathbb{N}$ to A by $f(i) = n_i$ for all $i \in \mathbb{N}$ and thus A is countably infinite. □

Theorem 1.7. *$f : A \to \mathbb{N}$ injective then $f : A \to f(A)$ is a bijection and thus A is countable.*

Proof.
- If $f(A)$ finite then A is finite and thus countable.
- If $f(A)$ infinite then there exists a bijection from $f(A)$ to $\mathbb{N}$ and thus we can define a bijection from A to $\mathbb{N}$ by $g(a) = h(f(a))$ where $h : f(A) \to \mathbb{N}$ is a bijection and thus A is countably infinite.

□

Corollary 1.3. *$\mathbb{N} \times \mathbb{N}$ is countable.*

Proof. $\mathbb{N} \times \mathbb{N} = \{(1,1),(1,2),(2,1),(1,3),(2,2),(3,1),...\}$ and we can define a bijection from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$ by $f(a,b) = 2^a 3^b$ for all $a, b \in \mathbb{N}$. □

Corollary 1.4. *If A is countable and $f : A \to B$ injective then A is countable.*

Proof. B countable then $\exists g : B \to \mathbb{N}$ bijection then $g \circ f : A \times B \to \mathbb{N}$ is an injection and thus $A \times B$ is countable. □

Corollary 1.5. *If A is countable and B countable then $A \times B$ is countable.*

Proof. A countable then $\exists f : A \to \mathbb{N}$ injective and B countable then $\exists g : B \to \mathbb{N}$ injective then $h : A \times B \to \mathbb{N} \times \mathbb{N}$ defined by $h(a,b) = (f(a), g(b))$ is injective and thus $A \times B$ is countable. □

Remark 1.3. $\{0,1\} \times \{0,1\} \times \dots$ is uncountable.

Proof. Assume it is then there exists a bijection from $\{0,1\} \times \{0,1\} \times \dots$ to $\mathbb{N}$ then we can list all the elements of $\{0,1\} \times \{0,1\} \times \dots$ as follows:

$$\begin{aligned}
&(a_{11}, a_{12}, a_{13}, \dots)\\
&(a_{21}, a_{22}, a_{23}, \dots)\\
&(a_{31}, a_{32}, a_{33}, \dots)\\
&\vdots
\end{aligned}$$

We can define an element of $\{0,1\} \times \{0,1\} \times \dots$ by $b_i = 1 - a_{ii}$ for all $i \in \mathbb{N}$ then $b = (b_1, b_2, b_3, \dots)$ is an element of $\{0,1\} \times \{0,1\} \times \dots$ but $b \neq (a_{11}, a_{12}, a_{13}, \dots)$ since $b_1 \neq a_{11}$, $b \neq (a_{21}, a_{22}, a_{23}, \dots)$ since $b_2 \neq a_{22}$, and so on we get $b \neq (a_{n1}, a_{n2}, a_{n3}, \dots)$ for all $n \in \mathbb{N}$ and thus b is not in the list which is a contradiction since we assumed that there exists a bijection from $\{0,1\} \times \{0,1\} \times \dots$ to $\mathbb{N}$ and thus $\{0,1\} \times \{0,1\} \times \dots$ is uncountable. □

Proposition 1.7. *The union of countable sets is countable.*

Proof. Let $A_1, A_2, A_3, \dots$ be countable.

- Case A_i disjoint: We can define a bijection from $\bigcup_{i=1}^{\infty} A_i$ to $\mathbb{N} \times \mathbb{N}$ by $f(a) = (i,j)$ where a is the j-th element of A_i and thus $\bigcup_{i=1}^{\infty} A_i$ is countable.

- Case A_i not disjoint: We can define a bijection from $\bigcup_{i=1}^{\infty} A_i$ to $\mathbb{N} \times \mathbb{N}$ by $f(a) = (i, j)$ where a is the j-th element of A_i and i is the smallest index such that a is the j-th element of A_i and thus $\bigcup_{i=1}^{\infty} A_i$ is countable.

□

Chapter 2

The set of Real Numbers

Definition 2.1. Let $(S, <)$ be an ordered set and $E \subseteq S$:

- We say that E is bounded from above if and only if $\exists a \in S$ such that $x \leq a \quad \forall x \in E$
- We say that E is bounded from below if and only if $\exists a \in S$ such that $a \leq x \quad \forall x \in E$
- We say that E is bounded if and only if E is bounded from above and bounded from below

Definition 2.2. Let $(S, <)$ be an ordered set and $E \subseteq S$. We say that $a \in S$ is the least upper bound of E if and only if a is an upper bound of E and for every upper bound b of E we have $a \leq b$. We denote the least upper bound of E also known as the supremum by $\sup(E)$. Similarly, we say that $a \in S$ is the greatest lower bound of E if and only if a is a lower bound of E and for every lower bound b of E we have $b \leq a$. We denote the greatest lower bound of E also called the infimum by $\inf(E)$.

Example 2.1. (1) Let $S = \{1, 2, 3\}$ and $E = \{1, 2\}$ then $\sup(E) = 3$ and $\inf(E) = 1$.

(2) Let $E = \{5, 6, 7, 8, \dots\}$ then E isnt bounded from above: Suppose it is then there exists $n_0 \in \mathbb{N}$ such that $x \geq n_0 \quad \forall x \in E$ but$n_0 + 1 \in E$ contradiction, and $\inf(E) = 5$.

Remark 2.1.
- By the well ordering principle every nonempty subset E of $\mathbb{N}$ has an infimum and $\inf_{\mathbb{N}} E \in E$
- If $E \subseteq S$ bounded from above and $\exists x_0 \in E$ this is an upperbound then $\sup(E) = x_0$ and thus $\sup(E) \in E$.

Example 2.2. In $\mathbb{Q}$ the set $E = \{\frac{1}{n}, n \in \mathbb{N}\}$ is bounded from above, 1 is an upper bound then $1 \in E$ then $\sup(E) = 1$. Claim: $\inf(E) = 0$. 0 is a lower bound of E, $\frac{1}{n} \geq 0 \quad \forall n \in \mathbb{N}$. Let $\gamma > 0, \gamma \in \mathbb{Q}$ so $\gamma = \frac{a}{b} \quad a, b \in \mathbb{N}$

$$0 \leq \frac{1}{b+1} \leq \frac{a}{b}$$

and $\frac{1}{b+1} \in E$ then γ is not a lower bound of E and thus $\inf(E) = 0$.

Example 2.3. Let

$$E = \{x \in \mathbb{Q} : x^2 < 2\}.$$

Then $E \neq \emptyset$ since $1 \in E$, and E is bounded above (for example by 2, because $x \geq 2 \Rightarrow x^2 \geq 4 > 2$, so no such x lies in E).

Assume, for contradiction, that $p = \sup E$ exists in $\mathbb{Q}$.

- If $p^2 < 2$, define
$$q := p + \frac{2 - p^2}{p + 2} = \frac{2(p + 1)}{p + 2} \in \mathbb{Q}.$$
Then $q > p$, and
$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} < 0.$$
So $q^2 < 2$, hence $q \in E$, contradicting p being an upper bound.

- If $p^2 > 2$, define
$$r := p - \frac{p^2 - 2}{p + 2} = \frac{2(p + 1)}{p + 2} \in \mathbb{Q}.$$
Then $r < p$, and
$$r^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0,$$
so $r^2 > 2$. Since $p \geq 1$, we have $r > 0$. If $x \in E$ and $x \geq r$, then $x^2 \geq r^2 > 2$, impossible; hence $x < r$ for all $x \in E$. Thus r is an upper bound of E smaller than p, contradiction.

Therefore $p^2 = 2$. But no rational number satisfies this: if $p = \frac{a}{b}$ in lowest terms, then
$$a^2 = 2b^2,$$
so a is even, say $a = 2k$, and then $b^2 = 2k^2$, so b is even, contradicting $\gcd(a, b) = 1$.

Hence $\sup(E)$ does not exist in $\mathbb{Q}$.

Definition 2.3. Give an ordered set $(S, <)$:

- We say that S satisfies the least upper bound property (L.U.P) if and only if for every non-empty subset E of S that is bounded from above admits a supremum in E.
- We say that S satisfies the greatest lower bound property (G.L.P) if and only if for every non-empty subset E of S that is bounded from below admits an infimum in E.

Proposition 2.1. *$LUP \iff GLP$*

Proof. S satifies LUP. Let E be a nonempty subset of S that is bounded from below take $A = \{a \in S$ such that $aisalowerboundofE\}$.

- $A \neq \emptyset$ since E is bounded from below.
- A is bounded from above, take $x_0 \in E$ take $a \leq x_0 \quad \forall a \in A$ then x_0 is an upperbound of A by LUP we have $a_0 = \sup(A)$ exists. By definition: Given $a \in A$ we have $a \leq a_0$ and given $b < a_0$ there exists $a \in A$ such that $b < a \leq a_0$. We claim that $a_0 = \inf(E)$. a_0 is a lower bound of E: Given $x \in E$ we have $a \leq x \quad \forall a \in A$ and thus $a_0 \leq x$. Given $\gamma > 0, \gamma < a_0$ so $\gamma < a_0 = \sup(A)$ then there exists $a \in A$ such that $\gamma < a \leq a_0$ and thus γ is not a lower bound of E and thus $\inf(E) = a_0$.

□

Theorem 2.1. *There exists a smallest algebraic field containing $\mathbb{Q}$ that satisfies LUP and preserving the operations on $\mathbb{Q}$. Moreover it is unique up to "isomorphism"(bijection preserving the structure). We call this algebraic field the field of real numbers denoted by $\mathbb{R}$.*

Idea of Proof. We define a dedekind cut as a subset of $\mathbb{Q}$ safisfying:

(1) $X \neq \emptyset$ and $X \subseteq \mathbb{Q}$

(2) if $p \in X$ then $q \in X$ for all $q < p$

(3) if $p \in X$ then there exists $r \in X$ such that $p < r$

Example 2.4 (Examples of Cuts)**.**

- $\{p \in \mathbb{Q} : p < \frac{1}{2}\}$ is a cut
- $\{p \in \mathbb{Q} : p \leq 1\}$ is not a cut

We define the set of real numbers to be the set of all cuts and we write $x <_{\mathbb{R}} y \iff x \neq y$

$<_{\mathbb{R}}$ **is an order:** Let x, y, z be cuts.

- Transitivity: Assume $x <_{\mathbb{R}} y$ and $y <_{\mathbb{R}} z$ then there exists $p \in y$ such that $p \notin x$ and there exists $q \in z$ such that $q \notin y$. Since $p \in y$ and $q \notin y$ we have $p < q$ and thus there exists $r \in z$ such that $p < r < q$ and thus $r \notin x$ and thus $x <_{\mathbb{R}} z$.
- Trichotomy: Assume $x \neq y$ then there exists $p \in y$ such that $p \notin x$ or there exists $q \in x$ such that $q \notin y$. In the first case we have $x <_{\mathbb{R}} y$ and in the second case we have $y <_{\mathbb{R}} x$.

Claim: $\mathbb{R}$ **satisfies LUP:** Let E be a non-empty subset of $\mathbb{R}$ that is bounded from above. We claim that $\sup(E) = \bigcup_{x \in E} x$. Let $x_0 = \bigcup_{x \in E} x$. x_0 is an upper bound of E: Given $y \in E$ we have $y \subseteq x_0$ and thus $y <_{\mathbb{R}} x_0$. Given $\gamma > 0, \gamma < x_0$ so there exists $p \in x_0$ such that $\gamma < p$ and thus there exists $x \in E$ such that $p \in x$ and thus $\gamma < p \in x$ and thus γ is not an upper bound of E and thus $\sup(E) = x_0$. □

Remark 2.2. We extend the real line $\mathbb{R}$ and include $\pm\infty$ denoted by $\mathbb{R}^{\infty} = \mathbb{R} \cup \{-\infty, +\infty\}$ such that $\forall x \in \mathbb{R}$ we have $-\infty < x < +\infty$. If $E \subseteq \mathbb{R}$, $E \neq \emptyset$ is not bounded above or below in $\mathbb{R}$ and we write $\sup(E) = \infty$ and $\inf(E) = -\infty$.

Theorem 2.2 (Archimedean Property)**.** *For every $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ such that $x < n$.*

Proof. Suppose that $\mathbb{N}$ is bounded in $\mathbb{R}$ then let $x_0 = \sup(\mathbb{N})$ then $x_0 - 1$ is not an upper bound then there exists n_0 such that $n_0 > x_0 - 1 \implies n_0 + 1 > x_0 = \sup \mathbb{N}$ but $n_0 + 1 \in \mathbb{N}$ contradiction. □

Corollary 2.1. $\inf_{\mathbb{R}}\{\frac{1}{n} : n \in \mathbb{N}\} = 0$. *That is, $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$*

Proof. Let $\varepsilon > 0$ and we have$\sup \mathbb{N} = \infty$ then there exists $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon} \implies \frac{1}{N} < \varepsilon$ □

Corollary 2.2. *For every $x \in \mathbb{R}$ there exists exactly one $m \in \mathbb{Z}$ such that $m \leq x \leq m + 1$ we call m the floor of x and write $m = \lfloor x \rfloor$*

Proof. $E = \{m \in \mathbb{Z} : m \leq x\}$

- $E \neq \emptyset$: $x \in \mathbb{R}$ then by the archimedean property $\exists N \in \mathbb{N}$ such that $N > x \implies -N < x$ and $-N \in \mathbb{Z}$ then $-N \in E$
- E is bounded from above in $\mathbb{Z}$ by the lemma $\forall m \in E : m \leq x \leq N$. By an exercise, $\exists m_0 = \sup E \in E \implies m_0 \leq x$ but if $m_0 + 1 < x \implies m_0 + 1 \in E$ contradiction thus $m_0 + 1 \geq x$ and thus $m_0 \leq x \leq m_0 + 1$ and thus $m_0 = \lfloor x \rfloor$.

Now we prove uniqueness: Assume there exists $m_1 \in \mathbb{Z}$ such that $m_1 \leq x \leq m_1 + 1$ and $m_1 \neq m_0$ then either $m_1 < m_0$ or $m_1 > m_0$. If $m_1 < m_0$ then $m_1 + 1 \leq m_0 < x$ contradiction. If $m_1 > m_0$ then $m_0 + 1 \leq m_1 < x$ contradiction. Thus, there exists exactly one $m \in \mathbb{Z}$ such that $m \leq x \leq m + 1$. □

Corollary 2.3. $\mathbb{Q}$ *is dense in* $\mathbb{R}$. *That is, given* $x, y \in \mathbb{R}$ *such that* $x < y$ *there exists* $q \in \mathbb{Q}$ *such that* $x < q < y$.

Proof. $x < y$ then $y - x > 0$ so by the archimedean property there exists $N \in \mathbb{N}$ such that $y - x > \frac{1}{N} \implies Ny - Nx > 1$. Take $m = \lfloor Nx \rfloor + 1 \in \mathbb{Z}$ then $Nx < m < Ny$ and $x < \frac{m}{N} < y$. □

2.1 Intervals

Definition 2.4. Given an ordered set S an interval $I \in S$ is defined by the following property: $x < y \quad x, y \in S \implies z \in S \quad \forall x \leq z \leq y$. In $\mathbb{R}$:

- $(a, b) = \{x : a < x < b\}$ is and open and bounded interval
- $[a, b] = \{x : a \leq x \leq b\}$ is a closed and bounded interval
- $[a, \infty) = \{x : a \leq x\}$ is a closed and unbounded interval
- $(-\infty, a] = \{x : x \leq a\}$ is an open and unbounded interval
- $(a, \infty) = \{x : a < x\}$ is an open and unbounded interval
- $(-\infty, a) = \{x : x < a\}$ is a open and unbounded interval

Theorem 2.3 (Nested Interval Theorem)**.** *Given a sequence of nested closed and bounded intervals* $I_n = [a_n, b_n]$ *such that*

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq [a_3, b_3] \supseteq \dots$$

then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ *Why do we need boundedness? Consider* $I_n = [n, \infty)$ *then* $\bigcap_{n=1}^{\infty} I_n = \emptyset$. *Why closedness? Consider* $I_n = (0, \frac{1}{n})$ *then* $\bigcap_{n=1}^{\infty} I_n = \emptyset$.

Proof. $A = \{a_1, a_2, \dots\}$ and $B = \{b_1, b_2, \dots\}$ and $a_n < b_1$ for all $n \in \mathbb{N}$, $a_2 < b_1$ ans so on. That is, $a_n < b_m \quad \forall m, n$. Fix m, b_m is an upper bound of A so $sup(A) \leq b_m \quad \forall m$ so $\sup(A)$ is a lower bound of B so $\sup(A) \leq \inf(B)$

$$x \in [\sup(A), \inf(B)] \implies a_n \leq x \leq b_n \implies x \in [a_n, b_n] = I_n \quad \forall n \implies x \in \bigcap_{n=1}^{\infty} I_n \implies \bigcap_{n=1}^{\infty} I_n \neq \emptyset$$

□

Corollary 2.4. $\mathbb{R}$ *is uncountable.*

Proof. Assume Countable then $\mathbb{R} = \{x_1, x_2, \dots\}$. Ler I_1 be an interval that is closed and bounded not containing x_1. Take $I_2 \subseteq I_1$, a closed and bounded inteval not containing x_2. Take $I_3 \subseteq I_2$, a closed and bounded inteval not containing x_3, and so on. Construct $I_{n+1} \subseteq I_n$ be closed and bounded while not containing x_{n+1}. So

$$x_k \notin \bigcap_{n=1}^{\infty} I_n \quad \forall k \in \mathbb{N} \implies \bigcap_{n=1}^{\infty} I_n = \emptyset$$

contradiction. □

2.2 A Few Cool Inequalities

Definition 2.5.

- $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ is the absolute value of x.
- $|x+y| \leq |x| + |y|$ is the triangle inequality.
- if $x > 1$ then $(1+x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$ is the Bernoulli's inequality.

Proof of Bernoulli's inequality. We argue by induction on n. **Base case:** For $n = 1$ we have $(1+x)^1 = 1 + 1x$ so the base case holds. **Induction step:** Assume that $(1+x)^n \geq 1 + nx$ for all $x > -1$ and we want to show that $(1+x)^{n+1} \geq 1 + (n+1)x$ for all $x > -1$. We have $(1+x)^{n+1} = (1+x)^n(1+x) \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x$ since $x^2 \geq 0$ and thus the induction step holds and thus Bernoulli's inequality holds. □

Remark 2.3. $(b^n - a^n) = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1})$ for all $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$. If $0 < a < b$ then $a^n - b^n \leq n(b-a)b^{n-1}$ for all $n \in \mathbb{N}$.

Theorem 2.4. *Let $n \in \mathbb{N}$ and $y > 0$ then there exists a unique $x > 0$ such that $x^n = y$*

Proof. — proof left as an exercise. □

Definition 2.6. Let $n \in \mathbb{N}$ and $y > 0$ we denote $\sqrt[n]{y} = y^{\frac{1}{n}}$ the unique $x > 0$ such that $x^n = y$. We call $\sqrt[n]{y}$ the n-th root of y. We denote $\sqrt{y} = \sqrt[2]{y}$ the square root of y.

Definition 2.7. Let $m \in \mathbb{Z}, n \in N$ and $x > 0$ we define $x^{\frac{m}{n}} = (x^m)^{\frac{1}{n}}$ show that its well defined. We also have: $x^{r+s} = x^r x^s$ and $x^{rs} = (x^r)^s$, and $(xy)^r = x^r \cdot y^r$.

Theorem 2.5 (Arithmetic Geometric Mean Inequality)**.** *$x_1, x_2, \ldots, x_n > 0$ then $\sqrt[n]{x_1 x_2 \ldots x_n} \leq \frac{x_1 + x_2 + \cdots + x_n}{n}$ and equality holds if and only if $x_1 = x_2 = \cdots = x_n$, and $\frac{x_1 + x_2}{2} \geq 2\sqrt{x_1 x_2}$*

Theorem 2.6 (Cauchy Induction)**.** *$S \subseteq \mathbb{N}$ such that $2^n \in S$ then $n \in S \implies n - 1 \in S$.*

Chapter 3

Metric Spaces

Definition 3.1. Let X be a set. A metric on X is a map

$$d : X \times X \to \mathbb{R}$$

such that:

(1) $d(x, y) \geq 0 \quad \forall x, y \in X$ with equality if and only if $x = y$.

(2) $d(x, y) = d(y, x) \quad \forall x, y \in X$.

(3) $d(x, z) \leq d(x, y) + d(y, z) \quad \forall x, y, z \in X$.

We call (X, d) a metric space.

Example 3.1.

- $(\mathbb{R}, |\cdot|)$ is a metric space.
- $(\mathbb{R}^2, d_1)$ where $d_1((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|$ is a metric space.
- $(\mathbb{R}^2, d_2)$ where $d_2((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ is a metric space.
- $(\mathbb{R}^2, d_\infty)$ where $d_\infty((x_1, y_1), (x_2, y_2)) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$ is a metric space.
- (X, d) is a discrete metric space where $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ if $x \neq y$ is a metric space.

Definition 3.2. (X, d) metric space, $x_0 \in X$ and $\varepsilon > 0$ we define the $\varepsilon-$neighborhood of x_0 by $N_\varepsilon(x_0) = \{x \in X : d(x, x_0) < \varepsilon\}$.

Definition 3.3. (X, d) metric space, $A \subseteq X$ and $x_0 \in X$ we say that x_0 is an interior point of A if and only if there exists $\varepsilon > 0$ such that $N_\varepsilon(x_0) \subseteq A$. We say that A is an open set if and only if every point in A is an interior point of A,

Example 3.2.

- $[0, 1)$ in $\mathbb{R}$, $1/2$ is an interior point of A since $N_\varepsilon(1/2) \subseteq [0, 1)$ for $\varepsilon < 1/2$. 0 is not an interior point of A since $N_\varepsilon(0) \not\subseteq [0, 1)$ for all $\varepsilon > 0$. Thus, $[0, 1)$ is not an open set.
- $(0, 1)$ in $\mathbb{R}$, $1/2$ is an interior point of A since $N_\varepsilon(1/2) \subseteq (0, 1)$ for $\varepsilon < 1/2$. 0 is not an interior point of A since $N_\varepsilon(0) \not\subseteq (0, 1)$ for all $\varepsilon > 0$. 1 is not an interior point of A since $N_\varepsilon(1) \not\subseteq (0, 1)$ for all $\varepsilon > 0$. Thus, $(0, 1)$ is an open set.
- (X, d_{disc}) discrete metric space, every point is an interior point of A since $N_\varepsilon(x) = \{x\}$ for $\varepsilon < 1$ and thus every subset of X is an open set.

Proposition 3.1. *(X, d) metric space, $x_0 \in X$ and $\varepsilon > 0$ then $N_\varepsilon(x_0)$ is open in X.*

Proof. Let $x \in N_\varepsilon(x_0)$ then $d(x, x_0) < \varepsilon$ so $\delta = \varepsilon - d(x, x_0) > 0$ and we claim that $N_\delta(x) \subseteq N_\varepsilon(x_0)$: Let $y \in N_\delta(x)$ then $d(y, x) < \delta$ so by the triangle inequality we have $d(y, x_0) \leq d(y, x) + d(x, x_0) < \delta + d(x, x_0) = \varepsilon$ and thus $y \in N_\varepsilon(x_0)$ and thus $N_\delta(x) \subseteq N_\varepsilon(x_0)$ and thus $N_\varepsilon(x_0)$ is open in X. □

Proposition 3.2.
- *Arbitrary union of open sets is open*
- *Finite intersection of open sets is open*

Remark 3.1. Infinite intersection of open sets is not necessarily open: Consider $A_n = (-\frac{1}{n}, \frac{1}{n})$ then $\bigcap_{n=1}^{\infty} A_n = \{0\}$ which is not an open set.

Proof.
- Let $\{A_\alpha\}_{\alpha \in I}$ be a family of open sets then we claim that $\bigcup_{\alpha \in I} A_\alpha$ is open: Let $x \in \bigcup_{\alpha \in I} A_\alpha$ then there exists $\alpha_0 \in I$ such that $x \in A_{\alpha_0}$ since A_{α_0} is open there exists $\varepsilon > 0$ such that $N_\varepsilon(x) \subseteq A_{\alpha_0} \subseteq \bigcup_{\alpha \in I} A_\alpha$ and thus $\bigcup_{\alpha \in I} A_\alpha$ is open.
- Let $A_1, A_2, \ldots, A_n$ be open sets then we claim that $\bigcap_{i=1}^{n} A_i$ is open: Let $x \in \bigcap_{i=1}^{n} A_i$ then $x \in A_i$ for all $i = 1, 2, \ldots, n$ since A_i is open there exists $\varepsilon_i > 0$ such that $N_{\varepsilon_i}(x) \subseteq A_i$ for all $i = 1, 2, \ldots, n$ take $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n\}$ then $N_\varepsilon(x) \subseteq N_{\varepsilon_i}(x) \subseteq A_i$ for all $i = 1, 2, \ldots, n$ and thus $N_\varepsilon(x) \subseteq \bigcap_{i=1}^{n} A_i$ and thus $\bigcap_{i=1}^{n} A_i$ is open.

□

3.1 Closed Sets

Notation 3.1. The deleted neighborhood of x_0 is defined by $N_\varepsilon^*(x_0) = N_\varepsilon(x_0) \setminus \{x_0\}$.

Definition 3.4. (X, d) metric space, $E \subseteq X$. We say that x is a limit point of E if and only if $\forall \varepsilon > 0$ we have $N^*(x_0) \cap E \neq \emptyset$. We denote by E' the set of limit points of E. We say that E is cloased if and only if $E' \subseteq E$. $E \setminus E'$ is the set of isolated points of E.

Example 3.3. $E = (0, 1]$ in $(\mathbb{R}, |\cdot|)$. Find E' and determine if E is closed.
Proof:

- $0 \in E'$: for any $\varepsilon > 0$ the interval $(0, \varepsilon)$ contains points of E (e.g. $\varepsilon/2 \in (0, 1]$ for sufficiently small ε), so every deleted neighborhood of 0 meets E.
- If $x \in (0, 1)$ then $x \in E'$: for any $\varepsilon > 0$ the neighborhood $(x - \varepsilon, x + \varepsilon)$ contains points of E different from x (for instance $x \pm \min\{\varepsilon/2, (1 - x)/2, x/2\}$), so x is a limit point.
- $1 \in E'$: for any $\varepsilon > 0$ the point $1 - \min\{\varepsilon/2, 1/2\} \in (0, 1)$ lies in the deleted neighborhood of 1, so every deleted neighborhood of 1 meets E.
- If $x < 0$ or $x > 1$ then $x \notin E'$: take $\varepsilon = \min\{|x|/2, |x - 1|/2\} > 0$, then $N_\varepsilon(x) \cap E = \emptyset$.

Hence $E' = [0, 1]$. The closure $\overline{E} = E \cup E' = [0, 1]$. Since $0 \notin E$, $E' \not\subseteq E$ and therefore E is not closed. Note also $E \setminus E' = \varnothing$, so E has no isolated points.

Remark 3.2. In $(\mathbb{R}, |\cdot|)$ every interior point is a limit point.

Example 3.4.
- $E = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$ in $(\mathbb{R}, |\cdot|)$ then $E' = \{0\}$ and thus E is closed and $E \setminus E' = \{\frac{1}{n} : n \in \mathbb{N}\}$ is the set of isolated points of E.

- $E = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$ in $(\mathbb{R}, d_{disc})$ then $E' = \emptyset$ and thus E is closed and $E \setminus E' = E$ is the set of isolated points of E.

Theorem 3.1. *(X, d) metric space, $E \subseteq X$ then E closed $\iff$ E^c is open.*

Consequence 3.1.

- *$[a, b], [a, \infty), (-\infty, a]$ are closed sets in $\mathbb{R}$.*
- *Arbitrary intersection of closed sets is closed.*
- *Finite union of closed sets is closed.*

Proof. ($\implies$) Assume E is closed then $E' \subseteq E$. Let $x \in E^c \implies x \notin E$. Since $E' \subseteq E$, we have $x \notin E'$, so there exists $\varepsilon > 0$ such that $N_\varepsilon(x) \cap E' = \emptyset$. This implies $N_\varepsilon(x) \subseteq E^c$, and thus E^c is open.

($\impliedby$) Assume E^c is open. Let $x \in E'$. Since E^c is open, there exists $\varepsilon > 0$ such that $N_\varepsilon(x) \subseteq E^c$. However, since $x \in E'$, every deleted neighborhood of x intersects E, which contradicts the fact that $N_\varepsilon(x)$ is entirely contained in E^c. Therefore, our assumption must be false, and it follows that $x \in E$. Hence, $E' \subseteq E$, and thus E is closed. □

Definition 3.5. (X, d) metric space, $E \subseteq X$. We define the closure of E as $\overline{E} = E \cup E'$.

Example 3.5. $E = [0, 1]$ in $(\mathbb{R}, |\cdot|)$ then $E' = [0, 1]$ and thus $\overline{E} = E \cup E' = [0, 1]$.

Example 3.6. $E = \mathbb{N}$ in $(\mathbb{R}, |\cdot|)$ then $E' = \emptyset$ and thus $\overline{E} = E \cup E' = \mathbb{N}$.

Example 3.7. $E = \mathbb{Q}$ in $(\mathbb{R}, |\cdot|)$ then $E' = \mathbb{R}$ and thus $\overline{E} = E \cup E' = \mathbb{R}$.

Theorem 3.2. *$\overline{E}$ is closed, In fact, $\overline{E}$ is the smallest closed set containing E. That is, if F is a closed set such that $E \subseteq F$ then $\overline{E} \subseteq F$.*

Proof. We will show $\overline{E}^c$ is open. Let $x \in \overline{E}^c$. Then $x \notin \overline{E} = E \cup E'$. Since $x \notin E$, we have $x \in E^c$. Since $x \notin E'$, there exists $\varepsilon > 0$ such that $N_\varepsilon(x) \cap E' = \emptyset$. This implies $N_\varepsilon(x) \subseteq E^c$, and thus $\overline{E}^c$ is open. So, $\overline{E}$ is closed. Let F closed such that $E \subseteq F$ then F^c is open and thus $\overline{E}^c \subseteq F^c$ and thus $F \subseteq \overline{E}$ and thus $\overline{E} \subseteq F$. □

Corollary 3.1. *In a metric space X, $E \subseteq X$ is closed if and only if $E = \overline{E}$.*

Proof. ($\implies$) Assume E is closed then $E' \subseteq E$ and thus $\overline{E} = E \cup E' = E$.

($\impliedby$) Assume $E = \overline{E}$ then $\overline{E}$ is closed and thus E is closed. □

3.2 Compact Sets

Definition 3.6. (X, d) metric space, $E \subseteq X$ we say that E is compact if and only if for every family of open cover $\{A_\alpha\}_{\alpha \in I}$ such that $E \subseteq \bigcup_{\alpha \in I} A_\alpha$ there exists a finite subcover $\{A_{\alpha_1}, A_{\alpha_2}, \ldots, A_{\alpha_n}\}$ such that $E \subseteq \bigcup_{i=1}^n A_{\alpha_i}$.

Example 3.8. $(0, 1)$ is not compact: Consider the open cover $\{(0, \frac{1}{n}) : n \in \mathbb{N}\}$ then there is no finite subcover since $\bigcup_{i=1}^n (0, \frac{1}{n_i}) = (0, \frac{1}{\max\{n_1, n_2, \ldots, n_n\}})$ and thus does not cover $(0, 1)$.

Example 3.9. $[0, \infty)$ is not compact: Consider the open cover $\{(-n, n) : n \in \mathbb{N}\}$ then there is no finite subcover since $\bigcup_{i=1}^n (-n_i, n_i) = (-\max\{n_1, n_2, \ldots, n_n\}, \max\{n_1, n_2, \ldots, n_n\})$ and thus does not cover $[0, \infty)$.

Exercise 3.1. Prove that $[0, 1)$ is not compact.

Theorem 3.3. *$[a, b]$ is compact in $\mathbb{R}$.*

Proof. Suppose, to the contrary, that there is an open cover $\{U_\alpha\}_{\alpha\in A}$ of $[a, b]$ with no finite subcover. Set $I_0 = [a_0, b_0] = [a, b]$. Having chosen a closed interval $I_n = [a_n, b_n]$ with no finite subcover, let $m_n = \frac{a_n+b_n}{2}$. If $[a_n, m_n]$ had a finite subcover then together with finitely many sets covering $[m_n, b_n]$ we would get a finite subcover of I_n, so at least one of the halves $[a_n, m_n]$ or $[m_n, b_n]$ has no finite subcover. Choose I_{n+1} to be such a half. This produces a nested sequence of closed intervals

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$$

whose lengths tend to 0.

By the Nested Interval Theorem the intersection $\bigcap_{n=0}^{\infty} I_n$ is nonempty; since the lengths go to 0 the intersection is a single point $x \in [a, b]$. As $\{U_\alpha\}$ covers $[a, b]$, some U_{α_0} contains x; because U_{α_0} is open there exists $\varepsilon > 0$ with $(x - \varepsilon, x + \varepsilon) \subseteq U_{\alpha_0}$. Choose N so that the length of I_N is $< \varepsilon$. Then $I_N \subset (x - \varepsilon, x + \varepsilon) \subseteq U_{\alpha_0}$, so I_N is covered by the single open set U_{α_0}, a finite subcover—contradiction.

Therefore every open cover of $[a, b]$ admits a finite subcover, i.e. $[a, b]$ is compact. □

Proposition 3.3. *(X, d) metric space, $E \subseteq X$ is compact then E is closed.*

Proof. We will show that E^c is open. Let $x \in E^c$ then $x \notin E$. For each $y \in E$, there exists $\varepsilon_y > 0$ such that $N_{\varepsilon_y}(y) \cap N_{\varepsilon_y}(x) = \emptyset$. The family $\{N_{\varepsilon_y}(y)\}_{y\in E}$ is an open cover of E, so there exist $y_1, y_2, \ldots, y_n \in E$ such that $\{N_{\varepsilon_{y_i}}(y_i)\}_{i=1}^n$ is a finite subcover of E. Let $\varepsilon = \min\{\varepsilon_{y_1}, \varepsilon_{y_2}, \ldots, \varepsilon_{y_n}\}$ then $N_\varepsilon(x) \subseteq N_{\varepsilon_{y_i}}(x)$ for all $i = 1, 2, \ldots, n$ and thus $N_\varepsilon(x) \cap N_{\varepsilon_{y_i}}(y_i) = \emptyset$ for all $i = 1, 2, \ldots, n$ and thus $N_\varepsilon(x) \cap E = \emptyset$ and thus $N_\varepsilon(x) \subseteq E^c$ and thus E^c is open and thus E is closed. □

Remark 3.3. The converse is not true: Consider $E = [0, \infty)$ in $(\mathbb{R}, |\cdot|)$ then E is closed but not compact.

Definition 3.7. (x, d) metric space and $A \subseteq X$ we say that A is bounded if and only if $\exists r > 0$ such that $A \subseteq N_r(x_0)$ with $x_0 \in X$.

Proposition 3.4. *(X, d) metric space, $E \subseteq X$ is compact then E is bounded.*

Proof. We will show that E is bounded. Fix $x_0 \in X$. Suppose, for contradiction, that E is unbounded. Then for every $n \in \mathbb{N}$, there exists $x_n \in E$ such that $d(x_n, x_0) \geq n$. The family $\{N_n(x_0)\}_{n=1}^{\infty}$ is an open cover of E (since E is unbounded, it meets every such neighborhood). By compactness, there exists a finite subcover $\{N_{n_1}(x_0), N_{n_2}(x_0), \ldots, N_{n_k}(x_0)\}$ such that $E \subseteq \bigcup_{i=1}^k N_{n_i}(x_0)$. Let $r = \max\{n_1, n_2, \ldots, n_k\}$. Then $E \subseteq N_r(x_0)$, which contradicts the unboundedness of E. Therefore, E is bounded. □

Theorem 3.4. *(X, d) metric space, $E \subseteq X$ compact then E is closed and bounded.*

Remark 3.4. The converse is false.

Proposition 3.5. *(X, d) metric space, K compact and $E \subseteq K$ closed then E is compact.*

Proof. We will show that E is compact. Let $\{A_\alpha\}_{\alpha\in I}$ be an open cover of E. Since E is closed, $K \setminus E$ is open. The family $\{A_\alpha\}_{\alpha\in I} \cup \{K \setminus E\}$ is an open cover of K, so there exist $\alpha_1, \alpha_2, \ldots, \alpha_n \in I$ such that $\{A_{\alpha_i}\}_{i=1}^n \cup \{K \setminus E\}$ is a finite subcover of K. Since $E \subseteq K$, we have $E \subseteq (K \setminus E) \cup \bigcup_{i=1}^n A_{\alpha_i}$. However, E cannot be covered by $K \setminus E$, so it must be covered by $\bigcup_{i=1}^n A_{\alpha_i}$. Therefore, $\{A_{\alpha_i}\}_{i=1}^n$ is a finite subcover of E, and thus E is compact. □

Theorem 3.5 (Heine-Borel)**.** *In* $(\mathbb{R}, |\cdot|)$ $E \subseteq \mathbb{R}$ *is compact if and only if* E *is closed and bounded.*

Proof. ($\Longrightarrow$) true in any metric space. ($\Longleftarrow$) E bounded then $E \subseteq [a, b]$ for some a, b but $[a, b]$ compact then E closed thus E compact. □

Theorem 3.6 (Sequential Compactness)**.** (X, d) *metric space,* $K \subseteq X$ *compact if and only if* $\forall E \subseteq K$ *infinite there exists a limit point of* E *in* K*.*

Proof. ($\Longrightarrow$) Assume K is compact. Let $E \subseteq K$ be infinite. Suppose, for contradiction, that E has no limit point in K. Then every point of E is an isolated point, so for each $x \in E$, there exists $\varepsilon_x > 0$ such that $N_{\varepsilon_x}(x) \cap E = \{x\}$. The family $\{N_{\varepsilon_x}(x)\}_{x \in E}$ is an open cover of E, and since $E \subseteq K$, it is also an open cover of K. By compactness, there exists a finite subcover $\{N_{\varepsilon_{x_i}}(x_i)\}_{i=1}^{n}$ such that $K \subseteq \bigcup_{i=1}^{n} N_{\varepsilon_{x_i}}(x_i)$. However, this implies that E is covered by finitely many neighborhoods, which contradicts the fact that E is infinite. Therefore, there must be a limit point of E in K.

($\Longleftarrow$) Proof left as exercise for $(\mathbb{R}, |\cdot|)$ and the general case is in math 233 and 305. □

Corollary 3.2 (Bolzano Weierstrass)**.** *Every infinite set in* $(\mathbb{R}, |\cdot|)$ *has a limit point.*

Proof. E is bounded in $(\mathbb{R}, |\cdot|)$ then $E \subseteq [a, b]$ but $[a, b]$ compact and E infinite by sequential compactness there exists a limit point of E in $[a, b]$ and thus in $\mathbb{R}$. □

Chapter 4

Sequences

Definition 4.1. (X, d) metric space, A sequence is a map from an ordered countable set usually $\mathbb{N}$ or $\mathbb{N} \cup \{0\}$ to X.

$$x_n = x(n) : \mathbb{N} \to X$$

Example 4.1. $x_n = \frac{1}{n}$ is an explicit sequence in $(\mathbb{R}, |\cdot|)$. $y_1 = 1$, $y_2 = 2$, $y_{n+2} = x_{n+1} + x_n$ is an implicit sequence in $(\mathbb{R}, |\cdot|)$.

Definition 4.2. Let $\{x_{n_k}\}$ be a sequence in (X, d). We say that x_n converges to x_0 in X if and only if $\forall \varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $x_n \in N_\varepsilon(x_0)$ for all $n \geq N$. That is, $\forall \varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d(x_n, x_0) < \varepsilon$ for all $n \geq N$. We denote by $\lim_{n\to\infty} x_n = x_0$ or $x_n \to x_0$ as $n \to \infty$. If no such x_0 exists we say that x_n diverges.

Proposition 4.1. *In a metric space the limit of a sequence if it exists is unique.*

Proof. Suppose x_n converges to x_0 and y_0 then $\forall \varepsilon > 0$ there exists $N_1, N_2 \in \mathbb{N}$ such that $d(x_n, x_0) < \varepsilon/2$ for all $n \geq N_1$ and $d(x_n, y_0) < \varepsilon/2$ for all $n \geq N_2$. Take $N = \max\{N_1, N_2\}$ then for all $n \geq N$ we have $d(x_0, y_0) \leq d(x_0, x_n) + d(x_n, y_0) < \varepsilon/2 + \varepsilon/2 = \varepsilon$ and thus $d(x_0, y_0) = 0$ and thus $x_0 = y_0$. □

Example 4.2. $x_n = \frac{1}{n}$ converges to 0 in $(\mathbb{R}, |\cdot|)$ since $\forall \varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$ for all $n \geq N$.

Example 4.3. $a_n = (-1)^n$ diverges. Suppose a_n converges to a_0 then $\forall \varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|(-1)^n - a_0| < \varepsilon$ for all $n \geq N$. Take $\varepsilon = 1/2$ then for all $n \geq N$ we have $|(-1)^n - a_0| < 1/2$ so $a_0 - 1/2 < (-1)^n < a_0 + 1/2$ for all $n \geq N$. If $a_0 \geq 0$ then $-1 < a_0 - 1/2 < (-1)^n < a_0 + 1/2 < 3/2$ for all $n \geq N$ which is a contradiction since $(-1)^n = -1$ for infinitely many n. If $a_0 < 0$ then $-3/2 < a_0 - 1/2 < (-1)^n < a_0 + 1/2 < 1/2$ for all $n \geq N$ which is a contradiction since $(-1)^n = 1$ for infinitely many n. Thus, a_n diverges.

Remark 4.1. (X, d) metric space, $E \subseteq X$ closed if and only if $\forall \{x_n\} \subseteq E$ if $\{x_n\}$ converges to x_0 then $x_0 \in E$.

Proof. ($\implies$) Assume E is closed. Let $\{x_n\} \subseteq E$ such that $\{x_n\}$ converges to x_0. Suppose, for contradiction, that $x_0 \notin E$. Since E is closed, E^c is open. Thus, there exists $\varepsilon > 0$ such that $N_\varepsilon(x_0) \subseteq E^c$. However, since $\{x_n\}$ converges to x_0, there exists $N \in \mathbb{N}$ such that $x_n \in N_\varepsilon(x_0)$ for all $n \geq N$. This implies that $x_n \in E^c$ for all $n \geq N$, which contradicts the fact that $\{x_n\} \subseteq E$. Therefore, our assumption must be false, and it follows that $x_0 \in E$.

($\impliedby$) Assume $\forall \{x_n\} \subseteq E$ if $\{x_n\}$ converges to x_0 then $x_0 \in E$. Suppose, for contradiction, that E is not closed. Then there exists a limit point of E, say y, such that $y \notin E$. Since y is a limit point

of E, there exists a sequence $\{y_n\} \subseteq E$ such that $\{y_n\}$ converges to y. However, by our assumption, since $\{y_n\} \subseteq E$ and $\{y_n\}$ converges to y, it must be the case that $y \in E$, which contradicts our earlier statement that $y \notin E$. Therefore, our assumption must be false, and it follows that E is closed. □

Definition 4.3. A sequence a_n is bounded if and only if $E = \{a_n : n \in \mathbb{N}\}$ is bounded. In $(\mathbb{R}, |\cdot|)$ this is equivalent to $\exists M > 0$ such that $|a_n| < M$ for all $n \in \mathbb{N}$.

Proposition 4.2. *(X, d) metric space, a_n is a sequence in X if a_n converges to a_0 then a_n is bounded.*

Remark 4.2. the converse is not true: Consider $a_n = (-1)^n$ in $(\mathbb{R}, |\cdot|)$ then a_n is bounded but does not converge.

Proof. a_n convergent and write $p = \lim_{n\to\infty} a_n$. Take $\varepsilon = 1$ then there exists $N \in \mathbb{N}$ such that $a_n \in N_1(p)$ for all $n \geq N$. Let $R = \max\{d(a_1, p), d(a_2, p), \ldots, d(a_{N-1}, p), 1\}$ then $a_n \in N_R(p)$ for all $n \in \mathbb{N}$ and thus a_n is bounded. □

4.1 Algebraic Properties

Theorem 4.1. *Let a_n and b_n be sequences in $(\mathbb{R}, |\cdot|)$ such that $\lim_{n\to\infty} a_n = a_0$ and $\lim_{n\to\infty} b_n = b_0$ then:*

(1) $\lim_{n\to\infty}(a_n + b_n) = a_0 + b_0$

(2) $\lim_{n\to\infty}(a_n b_n) = a_0 b_0$

(3) If $b_n \neq 0$ for all $n \in \mathbb{N}$ and $b_0 \neq 0$ then $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a_0}{b_0}$

(4) ca_n converges to ca_0 for all $c \in \mathbb{R}$.

Proof. We prove each statement using the definition of convergence and standard inequalities.

(1) **Sum.** Let $\varepsilon > 0$. Since $a_n \to a_0$ and $b_n \to b_0$, there exist $N_1, N_2 \in \mathbb{N}$ such that

$$|a_n - a_0| < \frac{\varepsilon}{2} \quad (n \geq N_1), \qquad |b_n - b_0| < \frac{\varepsilon}{2} \quad (n \geq N_2).$$

For $n \geq N = \max\{N_1, N_2\}$,

$$|(a_n + b_n) - (a_0 + b_0)| \leq |a_n - a_0| + |b_n - b_0| < \varepsilon.$$

Hence $a_n + b_n \to a_0 + b_0$.

(2) **Product.** Since $a_n \to a_0$, the sequence (a_n) is bounded; so there exists $M > 0$ such that $|a_n| \leq M$ for all n. Let $\varepsilon > 0$. Choose N_1 such that $|a_n - a_0| < 1$ for all $n \geq N_1$, and choose N_2 such that

$$|b_n - b_0| < \frac{\varepsilon}{2(M + |a_0| + 1)} \quad \text{for all } n \geq N_2.$$

Then for $n \geq N = \max\{N_1, N_2\}$,

$$\begin{aligned} |a_n b_n - a_0 b_0| &= |a_n(b_n - b_0) + b_0(a_n - a_0)| \\ &\leq |a_n|\,|b_n - b_0| + |b_0|\,|a_n - a_0| \\ &\leq M|b_n - b_0| + |b_0|\,|a_n - a_0|. \end{aligned}$$

Also $|a_n| \leq |a_0| + 1$ for $n \geq N_1$, so

$$|a_n b_n - a_0 b_0| \leq (M + |b_0|) \max\{|a_n - a_0|, |b_n - b_0|\}.$$

With the chosen bounds this is $< \varepsilon$. Thus $a_n b_n \to a_0 b_0$.

(3) **Quotient.** Since $b_0 \neq 0$ and $b_n \to b_0$, there exists N_1 such that

$$|b_n - b_0| < \frac{|b_0|}{2} \quad (n \geq N_1).$$

Hence for $n \geq N_1$,

$$|b_n| \geq \frac{|b_0|}{2}.$$

Now

$$\frac{a_n}{b_n} - \frac{a_0}{b_0} = \frac{a_n b_0 - a_0 b_n}{b_n b_0} = \frac{b_0(a_n - a_0) + a_0(b_0 - b_n)}{b_n b_0}.$$

Therefore, for $n \geq N_1$,

$$\left|\frac{a_n}{b_n} - \frac{a_0}{b_0}\right| \leq \frac{|b_0|\,|a_n - a_0| + |a_0|\,|b_n - b_0|}{|b_n||b_0|} \leq \frac{2}{|b_0|}|a_n - a_0| + \frac{2|a_0|}{|b_0|^2}|b_n - b_0|.$$

Choose N_2 so that both $|a_n - a_0|$ and $|b_n - b_0|$ are small enough for the right-hand side to be $< \varepsilon$ for all $n \geq N_2$. Then

$$\frac{a_n}{b_n} \to \frac{a_0}{b_0}.$$

(4) **Scalar multiples.** Let $c \in \mathbb{R}$. If $c = 0$, then $ca_n \to 0 = ca_0$. If $c \neq 0$, let $\varepsilon > 0$. Since $a_n \to a_0$, there exists N such that

$$|a_n - a_0| < \frac{\varepsilon}{|c|} \quad (n \geq N).$$

Then for $n \geq N$,

$$|ca_n - ca_0| = |c|\,|a_n - a_0| < \varepsilon.$$

So $ca_n \to ca_0$.

□

Exercise 4.1. Find $\lim_{n\to\infty} \frac{2n+1}{n+5}$.

Proposition 4.3. *In $(\mathbb{R}, |\cdot|)$ if $a_n \to L$ then $|a_n| \to |L|$*

Remark 4.3. The converse if false.

Proof. $||x| - |y|| \leq |x - y|$ for all $x, y \in \mathbb{R}$. Let $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. Then for all $n \geq N$ we have $||a_n| - |L|| \leq |a_n - L| < \varepsilon$ and thus $|a_n| \to |L|$. □

Theorem 4.2 (Sandwich/Squeeze Theorem)**.** *Let a_n, b_n and c_n be sequences in $(\mathbb{R}, |\cdot|)$ such that $a_n \leq b_n \leq c_n$ for all $n \in \mathbb{N}$. If $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$ then $\lim_{n\to\infty} b_n = L$.*

Proof. Let $\varepsilon > 0$ then there exists $N_1, N_2 \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N_1$ and $|c_n - L| < \varepsilon$ for all $n \geq N_2$. Take $N = \max\{N_1, N_2\}$ then for all $n \geq N$ we have $L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon$ and thus $|b_n - L| < \varepsilon$ and thus $b_n \to L$. □

Exercise 4.2. Suppose a_n and b_n converge to A and B respectively:

- Assume $a_n < b_n$ show that $A \leq B$
- Assume $a_n \leq b_n$ show that $A \leq B$

Proposition 4.4. *In* $(\mathbb{R}, |\cdot|)$. $a_n \to 0 \iff |a_n| \to 0$

Proof. ($\Longrightarrow$) Done

($\Longleftarrow$) $-|a_n| \leq a_n \leq |a_n|$ for all $n \in \mathbb{N}$ and $-|a_n| \to 0$ and $|a_n| \to 0$ thus by squeeze theorem $a_n \to 0$. □

4.2 Infinite Limits

Definition 4.4. a_n sequence in $\mathbb{R}$ we say that $a + n$ diverges to ∞ if and only if $\forall M > 0$ there exists $N \in \mathbb{N}$ such that $a_n > M$ for all $n \geq N$. We say that a_n diverges to $-\infty$ if and only if $\forall M < 0$ there exists $N \in \mathbb{N}$ such that $a_n < M$ for all $n \geq N$.

Example 4.4. $r > 0$, $\lim_{n\to\infty} r^n = \infty$ Proof: Let $M > 0 \in \mathbb{R}$ and let $N \in \mathbb{N}$ such that $N > M^{\frac{1}{r}}$ by archimedean property such N exists then for all $n \geq N$ we have $n^r > N^r > M$ and thus $\lim_{n\to\infty} r^n = \infty$.

4.3 Algebraic Properties

Let a_n and b_n be sequences in $\mathbb{R}$:

(1) If $a_n \to +\infty$ and b_n is bounded from be below then $a_n + b_n \to +\infty$.

(2) If $a_n \to -\infty$ and b_n is bounded from above then $a_n b_n \to -\infty$.

Proof. Without Loss of Generality, we will prove the first statement. Leet $M \in \mathbb{R}$ and b_n bounded from below by α, i.e., $b_n \geq \alpha \quad \forall n \in \mathbb{N}$, and $a_n \to +\infty$ implies that $a_n > M - \alpha \quad \forall n \geq N$ Then $a_n + b_n > M - \alpha + \alpha = M$ for all $n \geq N$ and thus $a_n + b_n \to +\infty$. □

If $a_n \to \infty$ and $b_n \to \alpha$:

(1) $\alpha > 0$ then $a_n b_n \to \infty$

(2) $\alpha < 0$ then $a_n b_n \to -\infty$

If $a_n \to \pm\infty$ and $a_n \neq 0$ for all $n \in \mathbb{N}$ then $\frac{1}{a_n} \to 0$.

Proof. Without loss of generality assume that $a_n \to +\infty$. Let $\varepsilon > 0$ since $a_n \to \infty$, there exists $N \in \mathbb{N}$ such that $a_n > \frac{1}{\varepsilon}$ for all $n \geq N$. Then $\frac{1}{a_n} < \varepsilon$ for all $n \geq N$ and thus $\frac{1}{a_n} \to 0$. □

Assume $a_n > 0$ for all $n \in \mathbb{N}$ and $a_n \to 0$ then $\frac{1}{a_n} \to \infty$. Siimilarly, if $a_n < 0$ for all $n \in \mathbb{N}$ and $a_n \to 0$ then $\frac{1}{a_n} \to -\infty$.

Theorem 4.3 (Squeeze Infinite Case)**.** *(1) If $a_n \geq b_n$ and $b_n \to \infty$ then $a_n \to \infty$*

(2) If $a_n \leq b_n$ and $b_n \to -\infty$ then $a_n \to -\infty$

Proof. — Done □

4.4 Monotone Sequences in $\mathbb{R}$

Definition 4.5. We say that a_n is nondecreasing if and only if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$. We say that a_n is nonincreasing if and only if $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}$. We say a_n is monotonic if and only if a_n is nondecreasing or nonincreasing.

Example 4.5. $a_n = (\frac{1}{2})^n$ is nonincreasing since $\frac{a_{n+1}}{a_n} = \frac{1}{2} < 1$ for all $n \in \mathbb{N}$.

Example 4.6. $a_n = (2.4)^n$ is nondecreasing since $\frac{a_{n+1}}{a_n} = 2.4 > 1$ for all $n \in \mathbb{N}$

4.4.1 Geometric Sequences

Definition 4.6. A geometric sequence is a sequence of the form $a_n = x^n$ for some $r \in \mathbb{R}$.

- If $x > 1$ then a_n is increasing (nondecreasing) and $\sup a_n = \infty$.
- If $0 < x < 1$ then a_n is decreasing (nonincreasing) and $\inf a_n = 0$.
- If $x = 0$ or $x = 1$ then a_n is constant
- If $x = -1$ then a_n is not monotonic and $\sup a_n = 1$ and $\inf a_n = -1$.
- If $x < -1$ then a_n is not monotonic and $\sup a_n = \infty$ and $\inf a_n = -\infty$.
- If $-1 < x < 0$ then a_n is not monotonic and $\sup a_n = x^2$ and $\inf a_n = x$.

Theorem 4.4 (Monotone Convergence Theorem)**.** *Let a_n be a monotonic sequence in $\mathbb{R}$ then a_n converges if and only if a_n is bounded. Moreover, if a_n is increasing and bounded from above then it converges, similarly if a_n is decreasing and bounded from below then it converges.*

Proof. ($\implies$) always true
($\impliedby$) If a_n increasing and bounded above then $a_n \to \sup a_n$. Similary if a_n decreasing and bounded below then $a_n \to \inf a_n$.

Proof. Let $L = \sup a_n \in \mathbb{R}$ then $\forall \varepsilon > 0$ we have $L - \varepsilon$ is not an upper bound of a_n then $\exists N \in \mathbb{N}$ such that $L - \varepsilon < a_N \leq L$ and since a_n is increasing we have $L - \varepsilon < a_N \leq a_n \leq L$ for all $n \geq N$ and thus $a_n \to L$. Similary if a_n decreasing and bounded below then $a_n \to \inf a_n$. □

□

Corollary 4.1. *If a_n is nondecreasing then $a_n \to \sup a_n$ where $\sup a_n$ can be ∞. If a_n is nonincreasing then $a_n \to \inf a_n$ where $\inf a_n$ can be $-\infty$.*

Proof. In the case where a_n is bounded above then we are done. Otherwise, let $M \in \mathbb{R}$ then there exists $N \in \mathbb{N}$ such that $a_N > M$ since nondecreasing we get $a_N > a_n > M \quad \forall n \geq N$ □

Theorem 4.5 (Basic Limits)**.** .

(1) $\lim_{n\to\infty} \sqrt[n]{n} = 1$

(2) $\lim_{n\to\infty}(1 + \frac{1}{n})^n = e$

(3) $\lim_{n\to\infty} \frac{x^n}{n!} = 0$

(4) $\lim_{n\to\infty} \frac{n^r}{a^n} = 0$

Proof. — left as exercise □

Remark 4.4. Let n_k be a strictly increasing sequence in $\mathbb{N}$ that is $n_1 < n_2 < \cdots \implies n_k > k$ so $\lim_{n\to\infty} k = \infty$ and thus $\lim_{k\to\infty} n_k = \infty$.

4.5 Subsequences

Definition 4.7. (X, d) metric space, a_n sequence in X let $n_1 < n_2 < \dots$ in $\mathbb{N}$. We define the subsequence $b_k = a_{n_k}$. If b_k converges to $\alpha \in X$ we say that α is a subsequential limit.

Example 4.7. a_{2n} and a_{2n+1} are subsequences of a_n. In general, if $f : \mathbb{N} \to \mathbb{N}$ then $a_{f(n)}$ is a subsequence of a_n.

Theorem 4.6. *(x, d) metric space, $a_n \to \alpha$ if and only if every subsequence of a_n converges to α.*

Proof. ($\Longleftarrow$) a_n is a subsequence of itself so $a_n \to \alpha$. ($\Longrightarrow$) Let b_k be a subsequence of a_n then $\forall \varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $a_n \in N_\varepsilon(\alpha)$ for all $n \geq N$. Since n_k is strictly increasing, there exists $K \in \mathbb{N}$ such that $n_K \geq N$ then for all $k \geq K$ we have $b_k = a_{n_k} \in N_\varepsilon(\alpha)$ and thus $b_k \to \alpha$. □

Exercise 4.3. In $(\mathbb{R}, |\cdot|)$ if $a_n \to \pm\infty$ then for every subsequence $b_k = a_{n_k}$ we have $b_n \to \pm\infty$.

Example 4.8. $a_n = \sin(n)$ diverges. Assume it converges to a limit L then

$$\exists n_1 \in (\frac{\pi}{6}, \frac{\pi}{2})$$

in $\mathbb{N}$ such that

$$\frac{1}{2} \leq \sin(n_1) \leq 1$$

, and

$$\exists n_2 \in (\frac{\pi}{6} + 2\pi, \frac{\pi}{2} + 2\pi)$$

such that

$$\frac{1}{2} \leq \sin(n_2) \leq 1$$

. We continue this to construct $n_k \in \mathbb{N}$ such that

$$\frac{1}{2} \leq \sin(n_k) \leq 1$$

Since $\sin(n) \to L$ then $\sin(n_k) \to L$ taking

$$n_1 \in (\frac{\pi}{6} + \pi, \frac{\pi}{2} + \pi)$$

We can construct another subsequence such that

$$-1 \leq \sin(n_k) \leq -\frac{1}{2}$$

and thus $\sin(n_k) \to L$ but this is a contradiction since L cannot be both greater than or equal to $\frac{1}{2}$ and less than or equal to $-\frac{1}{2}$. Thus, a_n diverges.

4.5.1 Subsequences in $(\mathbb{R}, |\cdot|)$

Theorem 4.7 (Corollary to Bolzano Weierstrass)**.** *Every bounded sequence in* $(\mathbb{R}, |\cdot|)$ *has a convergent subsequence.*

Proof. $E = \{a_n : n \in \mathbb{N}\}$

(1) E is finite then by pigeon hole $\exists x \in E$ such that $a_n = x$ for infinitely many n i.e $\exists n_1 < n_2 < \dots$ such that $a_{n_k} = x$ so $a_{n_k} \to x$.

(2) E is infinite then by bolzano weierstrass there exists a limit point x such that $x \in E'$. Take $\varepsilon = 1$ then $N_1^*(x) \cap E \neq \emptyset$ so $\exists n_1$ such that

$$0 < |a_n - x| < 1$$

. Keep doing this to get a sequence n_k such that $n_1 < n_2 < \cdots < n_k < \dots$ such that

$$0 < |a_{n_k} - x| < \frac{1}{k}$$

and thus $a_{n_k} \to x$.

□

Proposition 4.5. *If* a_n *is not bounded from above then there exists a subsequence* a_{n_k} *such that* $a_{n_k} \to \infty$*. Similarly, if* a_n *is not bounded from below then there exists a subsequence* a_{n_k} *such that* $a_{n_k} \to -\infty$*.*

4.5.2 Limsup and Liminf

Definition 4.8. Let a_n be a sequence in $\mathbb{R}$, assume that a_n is bounded from above and construct the sequence b_n to be $b_n = \sup\{a_k : k \geq n\}$ then $\lim_{n\to\infty} b_n = \inf(\sup\{a_k : k \geq n\}) = \limsup_{n\to\infty} a_n$. When a_n is not bounded from above then $\limsup_{n\to\infty} a_n = \infty$. Similarly, if a_n is bounded from below then $\liminf_{n\to\infty} a_n = \sup(\inf\{a_k : k \geq n\})$ and if a_n is not bounded from below then $\liminf_{n\to\infty} a_n = -\infty$.

Remark 4.5. $\liminf$ and $\limsup$ always exist in $R \cup \{-\infty, \infty\}$.

Proposition 4.6 (Properties of Limsup and Liminf)**.** *If* $a_n \leq b_n$ *then* $\liminf a_n \leq limsupa_n \leq \liminf b_n \leq \limsup b_n$*.*

Example 4.9. $a_n = (-1)^n$ then $\limsup a_n = 1$ and $\liminf a_n = -1$.

Example 4.10. $a_n = n$ then $\limsup a_n = \infty$ and $\liminf a_n = \infty$.

Theorem 4.8. a_n *is a sequence in* $\mathbb{R}$ *and* $\alpha \in \mathbb{R} \cup \{-\infty, \infty\}$ *then* $\liminf_{n\to\infty} a_n \leq \alpha \leq \limsup_{n\to\infty} a_n$

Proof. Case $\limsup a_n = \infty$ trivial. Case $\limsup a_n = -\infty$ but $a_n \leq \sup\{a_n, a_{n+1}, \dots\}$ then by infinite squeeze $a_n \to -\infty$ then every subsequence goes to $-\infty$. □

Example 4.11. $a_n - 3 + (-1)^n$ then any α subsequential limit is less than 4 but $a_{2n} \to 4$ then by theorem $\limsup a_n = 4$.

Theorem 4.9. *Let* a_n *be a sequence in* $\mathbb{R}$ *and* α *be a subsequential limit, then* $\liminf a_n \leq \alpha \leq \limsup a_n$*.*

Proof. Let $b_n = \{a_k : k \geq n\}$ and $c_n = \inf\{a_k : k \geq n\}$ then $c_n \leq a_n \leq b_n$. α is a subsequential limit so $\lim_{n\to\infty} a_{n_k} = \alpha$ then $\limsup a_n = \lim b_n = \lim b_{n_k}$ and $\liminf a_n = \lim c_n = \lim c_{n_k}$ then by squeeze theorem $\alpha \leq \limsup a_n$ and $\alpha \geq \liminf a_n$. □

Theorem 4.10. *Let a_n be a sequence in $\mathbb{R}$ then there exists a subsequence a_{n_k} such that $\lim a_{n_k} = \limsup a_n$ and a subsequence a_{n_e} such that $\lim a_{n_e} = \liminf a_n$.*

Proof.

- Case $\limsup a_n = \infty$ then there exists a subsequence a_{n_k} such that $a_{n_k} > k$ for all $k \in \mathbb{N}$ and thus $a_{n_k} \to \infty$.
- Case $\limsup a_n = -\infty$ then there exists a subsequence a_{n_k} such that $a_{n_k} < -k$ for all $k \in \mathbb{N}$ and thus $a_{n_k} \to -\infty$.
- Case $\limsup a_n = L \in \mathbb{R}$ take $\varepsilon = 1$ then there exist N_1 such that $L - 1 < L \leq b_{N_1} = \sup\{a_k : k \geq n\} \leq L + 1$ then there exists $n_1 \geq N_1$ such that $a_{n_1} > L - 1$. Take $\varepsilon = \frac{1}{2}$ then there exist N_2 such that $L - \frac{1}{2} < b_{N_2} = \sup\{a_k : k \geq n\} \leq L + \frac{1}{2}$ then there exists $n_2 \geq N_2$ such that $a_{n_2} > L - \frac{1}{2}$ and thus we can construct a subsequence a_{n_k} such that $a_{n_k} > L - \frac{1}{k}$ for all $k \in \mathbb{N}$ and thus $a_{n_k} \to L$.

□

Corollary 4.2. *Let a_n be a sequence in $\mathbb{R}$ then $\lim a_n = \alpha$ if and only if $\limsup a_n = \liminf a_n = \alpha$.*

Proof. ($\implies$) a_n is a subsequence of itself so $\limsup a_n = \liminf a_n = \alpha$.

($\impliedby$) Let $b_n = \sup\{a_k : k \geq n\}$ and $c_n = \inf\{a_k : k \geq n\}$ then $c_n \leq a_n \leq b_n$ for all $n \in \mathbb{N}$ and $\lim b_n = \limsup a_n = \alpha$ and $\lim c_n = \liminf a_n = \alpha$ then by squeeze theorem $a_n \to \alpha$. □

Theorem 4.11 (Nice Inequality)**.** *Let a_n be a sequence in $\mathbb{R}$ such that $a_n > 0$ then*

$$0 \leq \liminf \frac{a_{n+1}}{a_n} \leq \liminf \sqrt[n]{a_n} \leq \limsup \sqrt[n]{a_n} \leq \limsup \frac{a_{n+1}}{a_n}$$

Consequence 4.1. *$a_n > 0$ and $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \alpha$ then $\lim_{n\to\infty} \sqrt[n]{a_n} = \alpha$.*

Exercise 4.4. Find the value of $\lim_{n\to\infty} \sqrt[n]{n!}$.

Exercise 4.5. Let $\alpha > 0$ and $x_1 \geq \sqrt{\alpha}$

(1) Show that $x_n \geq \sqrt{\alpha} \quad \forall n - \frac{x_n + \frac{\alpha}{x_n}}{2} \geq \sqrt{\alpha}$.

(2) Show that x_n is nonincreasing

(3) Show that x_n converges and find the limit.

4.6 Cauchy Sequences

Definition 4.9. (X, d) metric space, a_n sequence in X we say that a_n is cauchy if and only if $\forall \varepsilon > 0$ there exists N such that $d(a_n, a_m) \leq \varepsilon \quad \forall m, n \geq N$

Example 4.12. $a_n = \frac{1}{n}$ is cauchy since $|a_n - a_m| = |\frac{1}{n} - \frac{1}{m}| \leq \frac{1}{N} < \varepsilon$ for all $n, m \geq N$.

Example 4.13. $a_n = (-1)^n$ is not cauchy since $|a_{2n} - a_{2n+1}| = 2$ for all $n \in \mathbb{N}$ and thus a_n is not cauchy.

Theorem 4.12. *If a_n is convergent then a_n is cauchy.*

Proof. a_n converges and write $p = \lim_{n\to\infty} a_n$. Let $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $a_n \in N_{\frac{\varepsilon}{2}}(p)$ for all $n \geq N$. Then for all $m, n \geq N$ we have $d(a_n, a_m) \leq d(a_n, p) + d(p, a_m) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ and thus a_n is cauchy. □

Remark 4.6. The converse isnt true. Take q_n a sequence of rationals coverging to $\sqrt{2}$ in $\mathbb{R}$ then q_n is cauchy but it does not converge in $\mathbb{Q}$.

Definition 4.10. A metric space (X, d) is complete if and only if every cauchy sequence in X converges to a point in X.

Example 4.14. $\mathbb{Q}$ is not complete since q_n from the previous example is cauchy but does not converge in $\mathbb{Q}$.

Lemma 4.1. *Let X be a metric space and a_n be a sequence in X such that if a_n cauchy then a_n is bounded.*

Proof. Take $\varepsilon = 1$ since a_n is cauchy then there exists N such that $d(a_n, a_m) < 1$ for $n, m \geq N$, let $R = \max\{1, d(a_1, a_N), \ldots, d(a_{N-1}, a_N)\}$ then $a_n \in N_{R+1}(a_N)$ □

Lemma 4.2. *Let X be a metric space and a_n be a cauchy sequence in X if a_n has a convergent subsequence then a_n converges.*

Proof. a_n cauchy and a_{n_k} converges to x then for all $\varepsilon > 0$ there exists N such that $d(a_n, a_m) < \frac{\varepsilon}{2}$ for $n, m \geq N$ and $a_{n_k} \to x$ then there exists K_1 such that $d(a_{n_k}, x) < \frac{\varepsilon}{2}$ for $k \geq K_1$ take $n_k \to \infty$ then there exists K_2 such that $n_k \geq N$ for all $k \geq K_2$ take $K = \max\{K_1, K_2\}$ then for all $n \geq N$ and $k \geq K$ we have $d(a_n, x) \leq d(a_n, a_{n_k}) + d(a_{n_k}, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ and thus $a_n \to x$. □

Theorem 4.13. $(\mathbb{R}, |\cdot|)$ *is complete*

Proof. Let a_n be caucy in $(\mathbb{R}, |\cdot|)$ then a_n is bounded and by the corollary of Bolzano Weierstrass there exists a subsequence a_{n_k} such that a_{n_k} converges to some $x \in \mathbb{R}$ then by the previous lemma a_n converges to x and thus $(\mathbb{R}, |\cdot|)$ is complete. □

4.6.1 Contractions (Special Case of Cauchy)

Definition 4.11. a_n is a contraction if there exists $c \in (0, 1)$ such that $\forall n$ we have $d(a_{n+2}, a_{n+1}) \leq cd(a_{n+1}, a_n)$.

Theorem 4.14. *Every contraction is cauchy.*

Proof. Let a_n be a contraction then there exists $c \in (0, 1)$ such that $\forall n$ we have $d(a_{n+2}, a_{n+1}) \leq cd(a_{n+1}, a_n)$ then by induction we can show that $d(a_{n+k}, a_{n+k-1}) \leq c^k d(a_{n+1}, a_n)$ for all $k \in \mathbb{N}$ and thus for all $m > n$ we have $d(a_n, a_m) \leq d(a_n, a_{n+1}) + d(a_{n+1}, a_{n+2}) + \cdots + d(a_{m-1}, a_m) \leq d(a_n, a_{n+1})(1 + c + c^2 + \ldots) = \frac{d(a_n, a_{n+1})}{1-c}$ and thus a_n is cauchy. □

Chapter 5

Series

Definition 5.1. Let a_n be a sequence in $\mathbb{R}$ then we define the sequence of partial sums

$$S_n = a_1 + s_2 \cdots + a_n = \sum_{k=1}^{n} a_k$$

. If S_n conveges we say that

$$\sum_{k=1}^{\infty} a_k$$

converges else it diverges. In particular cases where $\lim S_n = \alpha \in \mathbb{R} \cup \{\pm\infty\}$ we write $\sum_{k=1}^{\infty} a_k = \alpha$. In other words,

$$\sum_{k=1}^{\infty} a_k = \lim S_n$$

Example 5.1.

$$\sum_{k=1}^{\infty} \frac{1}{n} = \infty \text{ since } \lim_{n\to\infty} S_n = \lim_{n\to\infty} (1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}) = \infty$$

In fact we will show that

$$\sum_{k=1}^{\infty} \frac{1}{n^p}$$

converges for $p \geq 2$ and diverges for $p \leq 1$.

Notation 5.1. $\sum a_n$ denotes a series with no index

Theorem 5.1 (Algebraic Properties of Series)**.** *Let a_n and b_n be sequences in $\mathbb{R}$ and $c \in \mathbb{R}$ then*

(1) If $\sum a_n$ and $\sum b_n$ converge then $\sum(a_n + b_n)$ converges and $\sum(a_n + b_n) = \sum a_n + \sum b_n$

(2) If $\sum a_n$ converges then $\sum ca_n$ converges and $\sum ca_n = c\sum a_n$

Proof. (1) Let $S_n = \sum_{k=1}^{n} a_k$ and $T_n = \sum_{k=1}^{n} b_k$ then $S_n \to S$ and $T_n \to T$ for some $S, T \in \mathbb{R}$ then $\sum(a_n + b_n) = \lim_{n\to\infty}(S_n + T_n) = S + T = \sum a_n + \sum b_n$.

(2) Let $S_n = \sum_{k=1}^{n} a_k$ then $S_n \to S$ for some $S \in \mathbb{R}$ then $\sum ca_n = \lim_{n\to\infty} cS_n = cS = c\sum a_n$. □

Theorem 5.2 (nth term test)**.** *If $\sum a_n$ converges then $a_n \to 0$.*

Proof. Let $a_n = S_n - s_{n-1}$ and $S_n \to S$ then $S_{n-1} \to S$ and thus $a_n = S_n - S_{n-1} \to 0$. □

5.1 Series with Positive Terms

Remark 5.1. Let $a_n \geq 0$ and define the partial sum $S_n = a_1 + \dots a_n$ then S_n is nondecreasing. So by MCT S_n converges then S_n bounded so $\sum a_n$ converges and if S_n diverges then $\lim S_n = \infty$.

Theorem 5.3 (Comparison Test)**.** *Let a_n and b_n be sequences in $\mathbb{R}$ such that $0 \leq a_n \leq b_n$ for all $n \in \mathbb{N}$ then if $\sum b_n$ converges then $\sum a_n$ converges and if $\sum a_n$ diverges then $\sum b_n$ diverges*

Proof. Let $s_n \leq t_n$ and t_n convergent then t_n bounded say $t_n \leq t$ then $s_n \leq t$ so s_n bounded above and by the remark we get $\sum a_n$ converges. If $s_n \leq t_n$ and t_n diverges then by remark we get that $\lim t_n = \infty$ and by squeeze we get $\lim s_n = \infty$. □

Example 5.2. — Go back to your calculus textbook.

Corollary 5.1.

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges for } p > 1 \text{ and diverges for } p \leq 1$$

Proof. If $p \leq 0$ then the summation diverges by nth term text. If $p > 0$ then $\frac{1}{n^p} > 0$ and decreasing then the summation converges if and only if

$$\sum_{n=0}^{\infty} 2^n \cdot \frac{1}{2^{np}} = \sum_{n=0}^{\infty} \left(\frac{1}{2^{p-1}} \right)^n$$

converges which only happens when $p > 1$. □

Theorem 5.4 (Absolute Convergence Test)**.** *Let a_n be a sequence in $\mathbb{R}$ then if $\sum |a_n|$ converges then $\sum a_n$ converges. Moreover, if $\sum |a_n|$ converges then $|\sum a_n| \leq \sum |a_n|$ in particular, if $\sum |a_n|$ converges then $\sum a_n$ converges absolutely.*

Proof.

$$0 \leq |a_n| - a_n \leq 2|a_n|$$

and $\sum 2|a_n|$ coverges by comparison test then by squeeze $\sum(|a_n| - a_n)$ converges and thus $\sum a_n = \sum |a_n| - \sum(|a_n| - a_n)$ converges and $|\sum a_n| \leq |\sum |a_n|| + |\sum(|a_n| - a_n)| = \sum |a_n|$. □